Erasing Empires
Why Some Imperial Legacies Fade While Others Endure

Bogdan G. Popescu *

Abstract

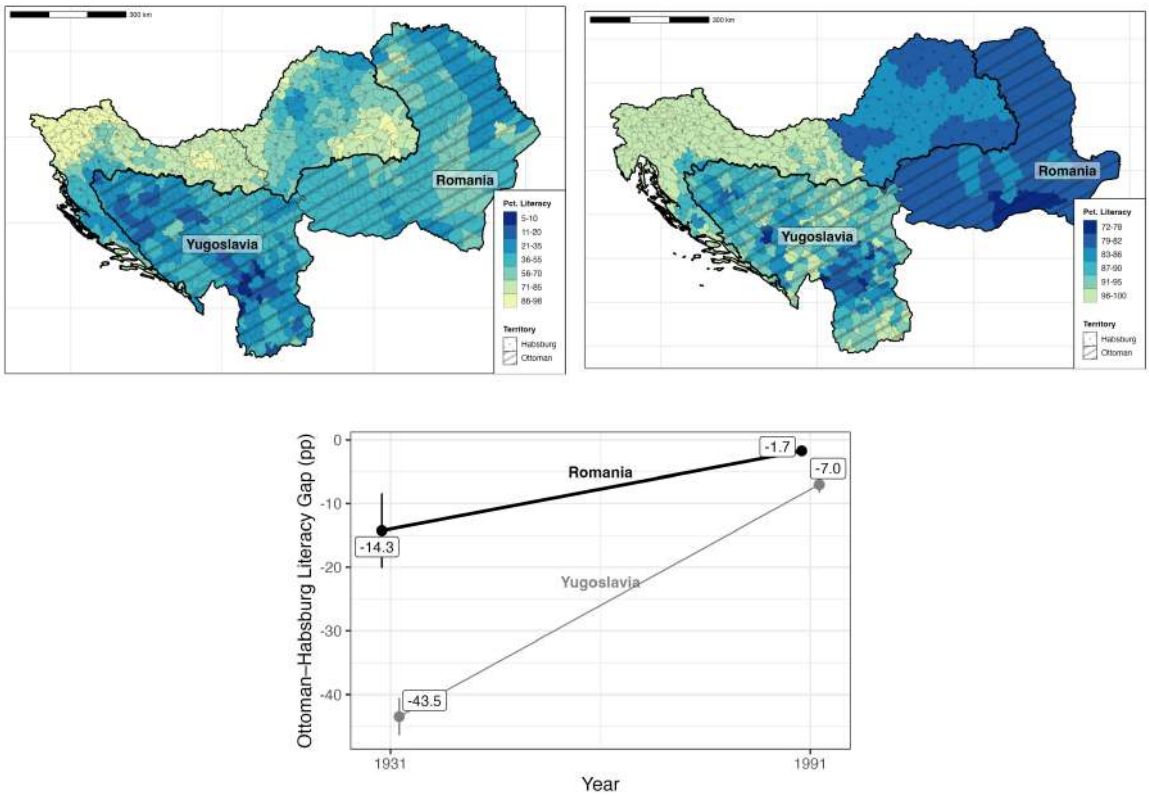
When do historical inequalities persist, and when do they fade? The Habsburg–Ottoman frontier produced literacy gaps of up to 40 percentage points in early twentieth-century Southeast Europe. Yet by 1991, these gaps had nearly vanished in Romania while remaining substantial in Yugoslavia's successor states. I argue that imperial legacies are conditionally erasable: convergence depends on the interaction between inherited administrative capacity and state–society alignment. Where centralized regimes govern ethnoreligiously aligned populations, they can overcome weak bureaucratic inheritances through mass campaigns. Where they confront status-reversed minorities misaligned with successor states, historical inequalities endure despite similar formal capacity. Using novel boundary discontinuity methods along historically credible segments of the 1739 Habsburg–Ottoman frontier, I demonstrate that Ottoman rule caused substantial literacy disadvantages, particularly among women. Communist schooling campaigns narrowed these gaps, but convergence was far more complete in homogeneous Romania (83% closure in log-odds) than in heterogeneous Yugoslavia (27% closure). Within Yugoslavia, literacy gaps persisted longest in Muslim-majority districts, where imperial status reversal fostered institutional mistrust-even as the negative effects of religious fractionalization attenuated over time. These findings specify scope conditions for institutional persistence and clarify when authoritarian state-building can rewrite historical hierarchies.

*Assistant Professor of Political Science, John Cabot University.
Email: bogdan.popescu@johncabot.edu

1 Introduction

When do historical inequalities persist, and when do they fade? Along the former Habsburg-Ottoman frontier in Southeast Europe, literacy rates in 1931 differed by as much as 40 percentage points on either side of the imperial boundary. Yet by 1991, these gaps had nearly disappeared in Romania while remaining substantial in Yugoslavia's successor states. Figure 1 illustrates this divergence. Both countries inherited similar administrative divides and both pursued centralized communist schooling after 1945. Why did mass education erase an imperial legacy in one setting but not the other?

Figure 1. *Romania and Yugoslavia Compared - Literacy in 1931 and 1991 overlaid with the Ottoman-Habsburg Border*



This article argues that centralized state capacity erodes historical inequalities

only under a specific social condition: state–society alignment. I define alignment as a relational condition in which local communities perceive core state institutions - especially public schools - as politically legitimate and identity-congruent. Where such alignment exists, uniform schooling policies translate into compliance, uptake, and rapid convergence. Where states govern historically misaligned populations - communities for whom imperial collapse produced a status reversal and whose identities became associated with a rival national project - the same formal capacity yields diminishing returns. In these settings, inherited inequalities persist not because states fail to invest, but because implementation and participation are costlier and more contested.

This argument speaks to two influential perspectives on long-run inequality. One emphasizes durable persistence: colonial and imperial institutions generate path-dependent differences in bureaucratic capacity and public goods provision that remain visible decades later (Acemoğlu *et al.*, 2001; Dell, 2010; Becker *et al.*, 2016). Another emphasizes erosion: centralized regimes, including authoritarian modernizers, can overcome historical disadvantages through mass schooling and policy standardization (Darden & Grzymała-Busse, 2006; Paglayan, 2022, 2024). The Romania-Yugoslavia divergence highlights what remains under-specified in both accounts: when does centralized authority translate into effective implementation across heterogeneous societies?

I develop a theory of *conditional erosion* in which the effectiveness of centralized state-building depends on local alignment. Building on work on infrastructural power and state penetration (Mann, 1993; Migdal, 1988), the theory links alignment to the costs of monitoring, enforcement, and participation. In aligned settings, shared identities and political incorporation reduce coordination and compliance costs, allowing states to equalize outcomes even where administrative lega-

cies were weak. In misaligned settings, diversity that maps onto historically status-reversed minorities raises monitoring costs and lowers voluntary compliance, making it harder for states to convert investment into schooling and literacy gains (Lieberman, 2009; Soifer, 2015; Scott, 1998).

I test this framework on the Habsburg-Ottoman frontier in Romania and in Yugoslavia's successor states. The frontier provides sharp subnational variation in imperial governance: the Habsburg Empire invested heavily in mass education and bureaucratic administration, whereas Ottoman governance relied more on indirect rule through local intermediaries and comparatively weaker state infrastructure in the periphery (Barkey, 2008; Becker *et al.*, 2016). The analysis proceeds in two steps. First, I estimate the Ottoman disadvantage in 1931 literacy using boundary-based RD2D methods that recover local treatment effects at multiple frontier points (Cattaneo *et al.*, 2025b,a). Second, I trace convergence under communist rule (1945–1991) to test whether erosion varied systematically with social structure.

The findings imply that institutional persistence is conditional, not uniform: the shadow cast by empires depends not only on inherited bureaucratic capacity, but also on how successor regimes govern populations whose political status and collective identities were transformed by the transition from imperial rule to national statehood. Methodologically, the paper also shows how historically grounded geographic RD designs can be implemented transparently by combining boundary-based estimation with documentary evidence on where local comparisons are credible.

The next section develops the theory of conditional erosion and derives testable implications. Section 3 describes the historical setting, data, and identification strategy. Section 4 presents the geographic RD estimates of Ottoman legacies in 1931. Section 5 examines convergence trajectories and mechanisms using within-Yugoslavia

variation and survey evidence. Section 6 concludes.

2 Argument

This paper advances a simple claim: imperial legacies are neither permanently fixed nor automatically erased by successor regimes. Instead, they are conditionally erasable: they decay only when state capacity and social alignment jointly permit sustained compliance with successor-state policies. Where regimes inherit weak bureaucracies but govern aligned populations, they can erode historical gaps. Where they confront status-reversed minorities misaligned with state institutions, legacies endure despite similar formal capacity.

This framework departs from existing work in two ways. First, it moves beyond the contrast between a *persistence view*, which emphasizes durable differences in state capacity and development (Acemoğlu *et al.*, 2001; Iyer, 2010; Dell, 2010; Dimitrova & Ramasheuskaya, 2021), and an *erasure view*, which highlights how centralized authoritarian regimes can overcome inherited disadvantages through mass campaigns and coercive mobilization (Paglayan, 2022, 2024). Second, it shifts attention from state capacity alone to a relational mechanism: *state–minority alignment*, rooted in the political reordering produced by imperial collapse, when formerly dominant or autonomous communities became minorities within new national states. Strong states can build schools everywhere; they cannot everywhere elicit the same willingness to comply.

Legacies vs. Mechanisms

I distinguish between *legacies* - durable structural differences inherited from imperial rule-and *mechanisms* - the processes through which these legacies shaped

outcomes under post-imperial regimes.

Two imperial legacies are central. First, the Habsburg and Ottoman empires left behind different *administrative infrastructures* that were visible in the territories that later became Romania and Yugoslavia. Habsburg rule generated professionalized, merit-based bureaucracies with standardized schooling systems (Becker *et al.*, 2016; Ingrao, 2000; Judson, 2016; Vogler, 2019), whereas late Ottoman rule relied more heavily on patrimonial arrangements and indirect governance, producing thinner and more fragmented local administration (Barkey, 1996; İnalcık, 1994, 1996). Second, Ottoman confessional governance through the millet system[1] created a specific *social structure*: religiously diverse populations in which Muslims were the ruling group aligned with imperial institutions. These two legacies - administrative infrastructure and social structure - emerged in parallel from Ottoman confessional politics.

Under communism, these legacies operated through two mechanisms. Inherited *bureaucratic capacity* shaped the resources and routines available for implementing mass schooling campaigns. Inherited *social structure*, especially the post-independence inversion of Muslim political status[2], shaped *state–minority alignment*: whether communities perceived successor institutions as legitimate or as vehicles of domination. Early in the communist period, religious diversity also generated *coordination costs* - disputes over language, script, and religious curricula - that slowed standardization (Bringa, 1995; Gažáková, 2014). These disputes were largely resolved by the 1960s, when coercive standardization made curricula, scripts, and

[1]The millet system was the Ottoman Empire's confessional governance regime, under which legally recognized religious communities (e.g. Muslims, Orthodox Christians, Armenians, and Jews) exercised autonomy over matters such as family law, education, and communal taxation. This institutional structure linked political status and administrative authority to religious identity and generated durable patterns of social segmentation across imperial territories.

[2]Independence inverted the status of Muslims in former Ottoman lands, turning an imperial ruling group into post-imperial minorities governed by Christian-dominated national states (Bringa, 1995; Donia & Fine, 1994).

compulsory schooling more uniform. Alignment, however, remained the persistent friction.

Table 1 summarizes the temporal sequence of legacies and mechanisms across the two imperial inheritances.

Table 1. *Temporal Sequence: Legacies and Mechanisms*

Period	**Ottoman Territory**	**Habsburg Territory**
Pre-1878 (Imperial legacies)	Millet system; thin bureaucracy; Muslims as ruling group	Merit-based bureaucracy; standardized schools
1878–1945 (Post-imperial)	Muslim status inverted; thin bureaucracy persists	Dense administration inherited
1945–1960s (Early communist)	Centralized schooling; alignment low in many Muslim areas	Centralized schooling; higher alignment
1960s–1991 (Late communist)	Coordination largely resolved; alignment frictions persist	Continued convergence

The Conditional Erosion Framework

The framework formalizes the claim that imperial legacies are *conditionally erasable* depending on two interacting factors: inherited administrative capacity and state–minority alignment. High inherited capacity combined with high alignment generates the greatest erosion of historical gaps, whereas low capacity and low alignment produce persistence. In practice, inherited capacity helps explain where initial disparities were largest, while alignment is central to explaining where centralized state-building could most effectively close them. This emphasis on alignment builds on work showing that social heterogeneity and indirect incorporation raise the costs of fiscal extraction and administrative reach, producing uneven state penetration even absent explicit discrimination (Charnysh, 2022; Magiya, 2025).

This distinction between inherited administrative capacity and state–minority alignment clarifies why centralized authoritarian regimes do not uniformly erase

historical inequalities. Even under high coercive capacity, alignment conditions whether compliance is sustained rather than merely imposed. Where misalignment persists, states can mandate schooling but face higher absenteeism, evasion, or symbolic compliance, producing slower convergence despite similar formal policies, higher expenditure, and comparable legal mandates.

The alignment mechanism captures how imperial collapse reconfigured political identities and institutional trust. Under the Ottoman Empire, Muslim populations were culturally and institutionally aligned with the imperial center, occupying positions of political dominance (Barkey, 2008; İnalcık, 1994). Independence inverted this relationship: Muslims in the Balkans became minorities in new Christian national states. This *status reversal* fostered mistrust toward public institutions now identified with rival ethnoreligious projects (Bringa, 1995; Ramet, 1992).

Where state institutions were identified with a dominant ethnoreligious group, minorities often viewed schools as instruments of cultural domination rather than neutral public goods (Alesina & Ferrara, 2005; Gerring *et al.*, 2015). Bringa's ethnography documents how older Muslim generations in Bosnia perceived state schools as culturally distant and morally threatening, while mosque-based instruction anchored an alternative community (Bringa, 1995, p. 204). This cultural distance translated into lower enrollment, greater resistance to girls' schooling, and skepticism about the neutrality of state curricula. Consistent with evidence that weak states allocate resources away from communities perceived as fiscally marginal or politically distant (Charnysh, 2022), post-imperial regimes did not encounter uniform receptivity to standardized schooling.

In more homogeneous former Ottoman territories such as Romania, where the dominant religious group (Orthodox Christians) controlled both imperial-era communal institutions and post-independence state structures, no comparable status

reversal occurred. Alignment between citizens and the communist state remained higher, facilitating compliance with centralized schooling initiatives. The same formal administrative capacity could thus yield divergent outcomes across communities: literacy advanced fastest where state legitimacy was high, and lagged where mistrust amplified compliance and monitoring costs.

Diversity, Coordination, and Alignment

This argument distinguishes between *heterogeneity* and *state–minority* misalignment. Heterogeneity denotes the coexistence of multiple ethnoreligious or linguistic groups within a territory. Misalignment refers to a relational condition in which a group associates state institutions with a dominant national project from which it feels excluded. These conditions impose distinct constraints on state policy.

Heterogeneity primarily generates coordination problems. In the contexnt of education, multiple languages, scripts, and educational traditions raise the fixed costs of implementing a uniform schooling system, but once a standard is selected and enforced, these costs decline as teachers, curricula, and cohorts are unified. State action can therefore resolve the constraint it initially confronts.

Misalignment, by contrast, generates a legitimacy problem. Where state institutions are identified with an alien or rival national project, schooling is normatively suspect. In such settings, expansion does not reduce future costs because it does not produce voluntary compliance: attendance can be mandated, but trust, parental investment, and perceived returns cannot be compelled. Resistance and symbolic compliance therefore persist even after formal standardization.

The distinction between heterogeneity and misalignment is thus dynamic rather than cultural. Coordination problems created by heterogeneity are self-resolving under centralization, whereas legitimacy problems created by misalignment are self-

reinforcing. Diversity alone may not impede convergence; only diversity that coincides with historical status reversal and weak institutional incorporation can produce durable limits to the erosion of imperial legacies.

Religious diversity created by the millet system in the Ottoman Empire initially generated coordination costs during the early communist period (1945–1960s). In Yugoslavia, authorities and Muslim religious leaders debated girls' schooling, coeducation, and the shape of Islamic instruction, with Bosnian Muslim associations acting as key interlocutors (Giomi, 2021). The legacy of Arebica (Arabic-based Bosnian script) illustrates script standardization frictions as the state promoted Latin and Cyrillic literacy (Gažáková, 2014).

These coordination costs were largely time-bound frictions rather than durable constraints. Communist regimes' coercive capacity and institutional standardization progressively resolved technical issues, and by the 1960s curricula, scripts, and compulsory schooling had become broadly uniform across both countries (Giomi, 2021). Yet literacy gaps persisted in Muslim-majority districts, indicating that alignment - not coordination complexity - remained the binding constraint once standardization was imposed.

The Yugoslav case illustrates a broader pattern identified in the literature on ethnic heterogeneity and public goods provision. Early work posited that ethnic diversity universally depresses public goods (Alesina *et al.*, 1999; Gisselquist, 2013), while more recent scholarship shows that outcomes depend on the type of good, level of analysis, and institutional context (Gerring *et al.*, 2015; Charnysh, 2019; Lee, 2018). An emerging insight is that negative correlations between heterogeneity and public goods often reflect the presence of historically marginalized groups rather than diversity as such. Studies of Brazil, India, and sub-Saharan Africa show that what appears to be a diversity effect is frequently driven by disadvantaged subpop-

ulations whose relationship with the state is shaped by exclusion or weak institutional incorporation (Gerring *et al.*, 2015; Kustov & Pardelli, 2018). This insight can be applied to post-imperial contexts by separating diversity as fractionalization and diversity as containing status-reversed minorities (e.g. Muslim population share).

Figure 2 summarizes the temporal sequence and causal pathways of the theoretical framework. Imperial legacies (squares) are structural inheritances: administrative infrastructure and social composition, largely determined during the imperial period. Communist-era policies started operating from these inherited structures. Mechanisms (solid circles) are processes taking place during 1945-1991: bureaucratic capacity and state–minority alignment. Coordination costs (circles) operated as a time-bound friction during the early communist period but attenuated as standardization progressed. The outcome (diamond) is literacy. The theory predicts that where bureaucratic capacity was initially low but alignment was high (as in relatively homogeneous Romania), centralized state-building could overcome administrative weakness, whereas where both administrative weakness and misalignment bound simultaneously (as in heterogeneous Yugoslavia), legacies persisted.

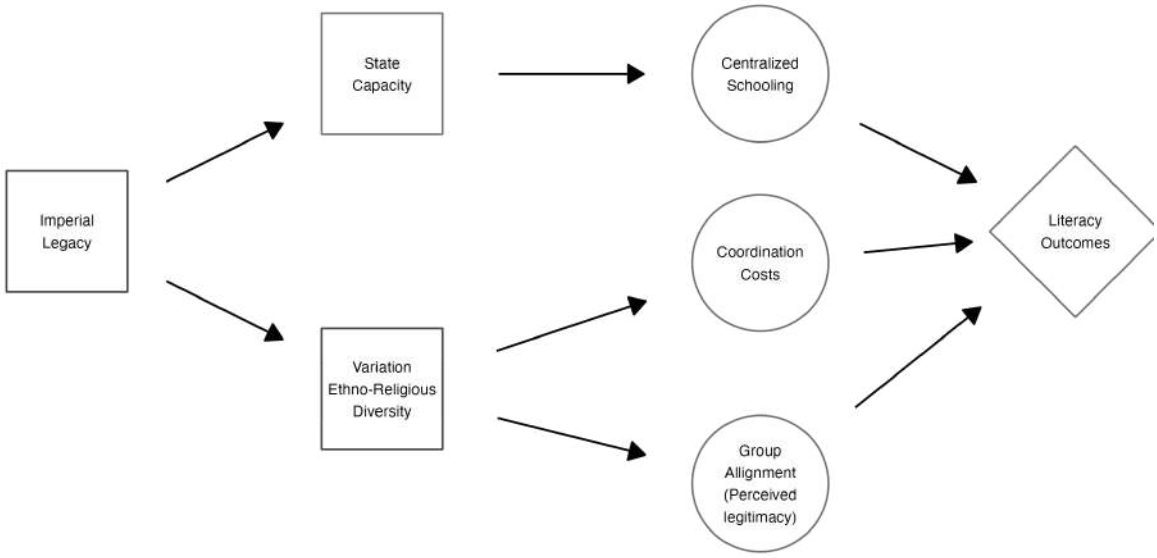
Observable Implications

These mechanisms yield a set of observable implications about the persistence and conditional erosion of imperial legacies in human capital.

H1 (Ottoman Legacy in Human Capital): *Localities historically governed by the Ottoman Empire will exhibit lower literacy rates in the early twentieth century, particularly among women.*

This reflects the transmission of weaker local bureaucracies from the late Ottoman period to successor states. Districts on the Ottoman side of the former frontier should also display lower contemporary indicators of state capacity, represent-

Figure 2. *Imperial Legacies and Communist-Era Mechanisms: Pathways to Literacy Outcomes*



Note: Squares represent imperial legacies (durable structural inheritances); solid circles represent persistent mechanisms - time-bound frictions (coordination costs, attenuating by the 1960s). The diamond is the outcome.

ing the Ottoman institutional inheritance.

H2 (Conditional Convergence and Temporal Dynamics): *Communist schooling campaigns will reduce historical literacy gaps, but convergence will be less complete in successor states where status-reversed minorities constitute a larger share of the population and remain misaligned with the national state.*

Where alignment is high (dominant-group continuity across imperial and post-imperial periods, as in Romania), communist state capacity can overcome inherited administrative weakness. Where status reversal created misaligned minorities (as in Yugoslavia), literacy gaps will persist despite centralized schooling campaigns and institutional standardization.

H3 (Alignment Mechanism): *Within former Ottoman territories, literacy gaps will persist longest in districts with larger Muslim populations.*[3]

[3]Whereas H2 concerns cross-national convergence trajectories, H3 isolates the alignment mechanism through within-country variation.

The mechanism predicts that Muslim concentration - rather than heterogeneity per se - will be the persistent correlate of educational gaps within successor states, conditional on inherited administrative capacity, geography, and baseline development.

3 Historical Context: Empires, Bureaucracy, and Schooling

The Ottoman administrative order classified subjects by religion and organized communal affairs-especially family law and schooling-through recognized religious communities (millets), notably the Rum (Greek Orthodox), Armenian, and Jewish communities (Barkey, 2008; İnalcık, 1973). While often portrayed as tolerant, this system institutionalized difference: communal authorities managed their own laws, and fiscal obligations varied by confession (Braude & Lewis, 1982; Faroqhi, 2005). Religious identity thus shaped how subjects interacted with imperial institutions.

Before the nineteenth century, the limited education in the Ottoman Empire was decentralized and primarily religious. Society was divided between the ruling estate - political, military, and religious elites - and the tax-paying subjects (*reaya*); educational provision reflected this hierarchy (İnalcık, 1994). Elementary instruction for Muslims took place in Qur'anic schools attached to mosques or local endowments (*vakıf*), where children-boys and sometimes girls-learned basic literacy, arithmetic, and religious recitation (Barkey, 2008; Fortna, 2016; Somel, 1997). Advanced study occurred in *medreses*, which trained the religious–legal elite and supplied bureaucratic personnel (Findley, 1996). These institutions were locally funded and unevenly distributed-dense in urban centers such as Istanbul, Bursa, and Edirne but sparse in peripheral regions. Parallel Christian and Jewish school networks existed under monasteries or rabbinical authorities, similarly funded by communal en-

dowments and oriented toward religious instruction (Sharkey, 2017). Thus, before the Tanzimat reforms (1839–1876), education was not a public good but a mosaic of confessional initiatives embedded in religious hierarchies.

During the nineteenth century, the empire expanded secular-curriculum schools as part of the Tanzimat reforms (Fortna, 2016; Somel, 1997). The result was a dual educational landscape: centrally supervised schools clustered around administrative centers, while communal schools predominated elsewhere. These reforms, however, rested on a fragile administrative base. From the late seventeenth century the Ottoman state shifted away from the classical land-military system[4] toward tax-farming contracts[5], which eroded central oversight and fragmented governance (İnalcık, 1996). The state increasingly relied on local intermediaries for revenue and order, producing uneven administrative penetration (Findley, 1996).

Over roughly the same period, the Habsburgs moved in the opposite direction-toward bureaucratic integration. The Theresian and Josephinian reforms professionalized the civil service, expanded cadastral and fiscal registration, and standardized schooling across ethnolinguistic regions (Ingrao, 2000; Judson, 2016). This yielded a merit-based, records-rich, and centrally supervised apparatus that successor states could later deploy for mass education (Becker *et al.*, 2016; Vogler, 2019, 2023).

Educational policy was central to this process. From Maria Theresa's 1774 *Allgemeine Schulordnung* onward, the monarchy pursued compulsory primary schooling as a tool of civic discipline and imperial cohesion (Deak, 2014; Judson, 2016). School inspectors, standardized curricula, and teacher training colleges extended state over-

[4]The classical Ottoman land–military (*timar*) system granted revenue rights to cavalry officers in exchange for military service, financing provincial forces without a centralized fiscal bureaucracy (İnalcık, 1994).

[5]Tax-farming (*iltizām*) auctioned tax-collection rights to private contractors, who kept surpluses and thereby gained local power at the expense of central oversight (Pamuk, 2001).

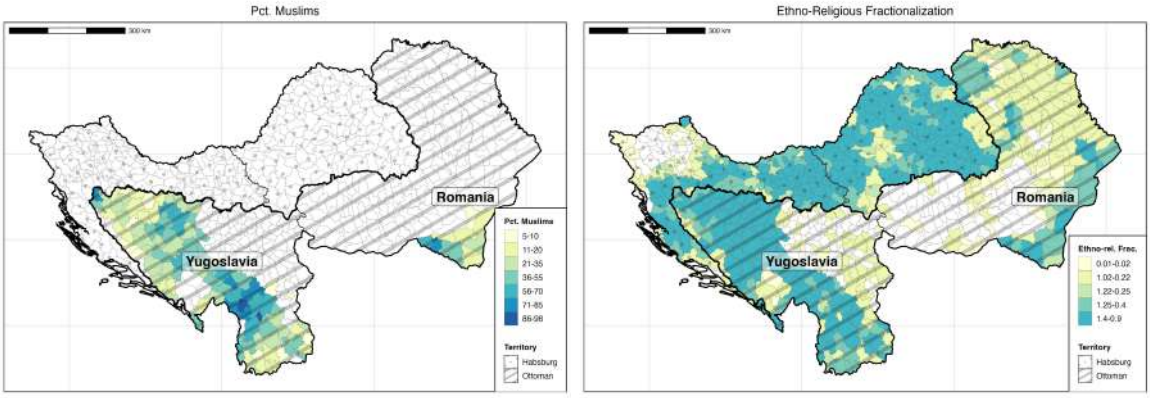
sight into rural parishes and multiethnic peripheries. By the mid-nineteenth century, education had become a bureaucratic domain-subsidized by the state (Bérenger, 2014; Cvrček, 2020; Komlos, 1983). These institutions fostered early mass literacy and habituated populations to regularized state–citizen contact. Whereas the Ottoman model relied on negotiated provision, the Habsburgs embedded schooling within a vertically integrated bureaucracy that reached from Vienna to the village schoolroom.

Romania and Yugoslavia exemplify this uneven imperial inheritance. Both encompassed territories formerly ruled by the Habsburg and Ottoman Empires and displayed substantial subnational variation in religious composition. Interwar censuses indicate that the Ottoman side of Romania was predominantly Orthodox and comparatively homogeneous, whereas the Ottoman side of Yugoslavia included Orthodox, Catholic, and Muslim communities. The Habsburg side of both countries, by contrast, was markedly more heterogeneous, home to intermingled Orthodox, Catholic, Protestant, and Jewish populations. This internal diversity allows for "like-with-like" comparisons: by holding national institutions constant, one can assess how inherited administrative density and local social structure jointly conditioned the uptake of standardized schooling.

Interwar censuses indicate that the Ottoman side of Romania was predominantly Orthodox and comparatively homogeneous, whereas the Ottoman side of Yugoslavia included Orthodox, Catholic, and Muslim communities. Figure 3 maps the distribution of muslims and ethnoreligious fractionalization throughout Romania and Yugoslavia in 1931. Muslims are much more present in regions like Bosnia (31%), Kosovo (72%), Macedonia (27%), and Montenegro (15%), and much less in Croatia (0.15), Romania (1%), or Serbia (2.8%). I measure ethnoreligious fractionalization through a standard district-level index capturing the probability that two randomly

selected residents belong to different religious groups.[6] Descriptively, the former Ottoman regions of Romania are substantially more religiously homogeneous (mean = 0.07) than those of Yugoslavia (mean = 0.24), as indicated in panel 2 of Figure 3.

Figure 3. *Romania and Yugoslavia Compared - Pct. Muslims and Ethno-Religious Fractionalization overlaid with the Ottoman-Habsburg Border*



4 Methodology

The empirical strategy proceeds in two stages with distinct identification goals. Stage 1 estimates the total causal effect of Ottoman rule on literacy in 1931 using a geographic regression discontinuity (GRD2D) design. Treatment is inherited imperial exposure: a district is coded as treated if its centroid lies on the Ottoman side of the frontier as configured by the Treaty of Belgrade (1739) - the most durable and institutionally salient demarcation of Ottoman and Habsburg rule - and untreated otherwise. As such, the RD identifies an intention-to-treat legacy effect of imperial rule at the frontier. Appendix Section 4 documents the construction of the historical boundary and the rationale for the 1739 cutoff. It also shows that alternative border configurations (1664, 1699, 1718) generate smaller discontinuities in Ottoman

[6]For 1931, I use six groups-Muslims, Orthodox, Catholics, Protestants, Jews, and others. For 1961, 1981, and 1991, the index is constructed using Muslims, Catholics, and Orthodox.

exposure at the cutoff, consistent with 1739 being the most institutionally salient boundary for RD identification.

The design exploits plausibly exogenous variation along historically credible segments of the Habsburg-Ottoman frontier. This approach deals with the problem that distance-to-boundary conditioning in traditional GRD can be biased near kinks and irregular borders (Cattaneo *et al.*, 2025a). The analysis draws on a newly harmonized dataset constructed from interwar and late socialist censuses of Romania and the component units of Yugoslavia (Bosnia and Herzegovina, Serbia, Croatia, Montenegro, North Macedonia, and Slovenia). For each census wave between 1930/31 and 1991, I assemble district-level measures of literacy (by gender), religious composition, and basic demographics; harmonize historical and contemporary boundaries; and georeference frontier segments. The unit of analysis in Stage 1 is the district at each census wave, with literacy measured as the share of the adult population able to read and write. The resulting panel spans six decades and seven successor states of the Habsburg and Ottoman empires.

Stage 2 examines heterogeneity in post-imperial convergence by testing whether the persistence of literacy gaps varied systematically with two inherited conditions: administrative capacity and ethnoreligious composition. Ottoman rule shaped both state capacity and social structure through confessional governance, and these legacies operated in parallel rather than sequentially. Accordingly, Stage 2 does not estimate causal mediation. Instead, it evaluates whether the total Ottoman effect identified in Stage 1 attenuated differently across social contexts under centralized socialist regimes. Specifically, I test whether convergence was faster where state–society alignment was stronger, conditional on initially limited bureaucratic capacity - predictions that follow directly from the theoretical framework. Although these conditions are not experimentally manipulated, systematic differences in the ero-

sion of literacy gaps provide discriminating evidence about the role of alignment in conditioning the persistence of imperial legacies.

The first stage uses the RD2D estimator along the Habsburg-Ottoman frontier to compare adjacent localities that share geography but differ in imperial exposure. Conceptually, the design follows the standard GRD logic of comparing outcomes just across a political boundary. Formally, however, this multidimensional boundary discontinuity framework of Cattaneo *et al.* (2025a) treats treatment assignment as a function of two-dimensional geographic location relative to the frontier rather than as a function of one-dimensional signed distance. Identification comes from discontinuities in outcomes as locations approach the boundary from either side in geographic space. In addition, the estimator is robust to irregular boundary geometry.

The primary estimand is the local boundary treatment effect of Ottoman rule on literacy at frontier location $b \in \mathscr{B}$. Formally, the causal parameter of interest is:

$$\tau(b) \equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = b], \tag{1}$$

where Y_i denotes district-level literacy, X_i denotes the geographic location of district i, and $\mathscr{B}$ denotes the Habsburg-Ottoman frontier. Under standard regression discontinuity continuity assumptions, $\tau(b)$ is identified as:

$$\tau(b) = \lim_{x \to b,\, x \in \mathscr{A}_1} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \to b,\, x \in \mathscr{A}_0} \mathbb{E}[Y_i \mid X_i = x], \tag{2}$$

where $\mathscr{A}_1$ and $\mathscr{A}_0$ denote the Habsburg and Ottoman sides of the frontier, respectively. Substantively, $\tau(b)$ represents the difference in adult literacy between districts on the Ottoman and Habsburg sides of the frontier in an arbitrarily small neighborhood around location b.

In addition to reporting the full distribution of $\tau(b)$ along the frontier, I distinguish between local effects within historically low-stakes and high-stakes segments of the boundary. Estimates from low-stakes segments constitute the primary causal estimands and anchor the causal interpretation of the analysis.

Within the low-stakes regions, the RD conditions are most plausible in segments where the frontier was fixed for reasons orthogonal to local development. As Kocher and Monteiro note, a valid natural experiment is one in which "a natural or social process has assigned units to distinct conditions in a way that is as good as random" (Kocher & Monteiro, 2016, p. 952). If the line were drawn to include systematically wealthier or more literate districts on one side, I could not "distinguish the effects of causes from the effects of the conditions under which they operate" (*ibid*). Assessing this assumption requires close, idiographic historical work on how the frontier was fixed.[7] Moreover, Paine *et al.* (2024, p. 17) argue that even when borders were not drawn in an as-if-random fashion overall, conditional randomness may be plausible for segments anchored to geographic features such as rivers.

Assignment to Treatment: Drawing of the Border

This section reconstructs treatment-assignment causal process observations. Following Kocher & Monteiro (2016), I assess the plausibility of as-if-random assignment by examining (i) the information available to negotiators, (ii) the incentives shaping border placement, and (iii) their capacity to manipulate the line.

(1) What information was available to the negotiators?

Before the late seventeenth century, European frontiers were often porous zones

[7] Two critiques illustrate the risks. Kocher & Monteiro (2016) re-examine Ferweda & Miller (2014) on the Vichy demarcation line, showing that the boundary followed double-track railways—a strategic choice that undermined comparability. Paine *et al.* (2024) highlight similar issues for African colonial borders, which often tracked rivers or pre-existing polities. Both conclude that GRDs are credible when researchers can demonstrate, using historical evidence, that the frontier was determined for reasons orthogonal to the outcome of interest.

rather than fixed lines. For the GRD, what matters is that the Habsburg-Ottoman boundary became a clearly delineated frontier after the Treaty of Karlowitz (1699), and durable in form after the Treaty of Belgrade (1739) (see appendix sextion 4 for an overview of border configurations over time). Roughly 70% of the 1739 line coincided with the 1699 demarcation.[8] Joint commissions of engineers and cartographers conducted on-the-ground surveys to document the physical and strategic features of the frontier. They mapped rivers, marshes, and ridgelines to identify defensible and stable boundaries, and recorded forts, villages, and trade routes. Surveyors also noted soil quality, road conditions, and access to water. Rivers and mountain ridges became anchors for the demarcation line (Fodor, 2009; Kurtaran, 2018; Gümüşçü, 2010). The three rivers that coincide with the border - Una, Sava, and the Danube - are visible in the first panel of Figure 4. As Stoye writes, "in war, or if war threatened, the river became a military frontier adjoining a communication system for the forces" (Stoye, 1994, p. 83). Since mass literacy emerged only later, negotiators were unlikely to have information about education outcomes when fixing the boundary.

(2) What incentives-military, political, or economic-shaped their choices? The commissions' information-gathering was guided by pragmatic military, political, and economic considerations, consistent with the principle of *uti possidetis*, whereby each side sought to retain territories it effectively occupied at the close of hostilities (Ingrao, 2000; Rothenberg, 1966; Shaw & Shaw, 1976; Stoye, 1994). Militarily, both empires prioritized defensible terrain - rivers, ridgelines, and marshes - that reduced the need for costly garrisons. Politically, they aimed to establish a recognized border to reduce skirmishes and stabilize diplomacy. Economically, sur-

[8] On the shift from jurisdictional to territorial sovereignty, see Febvre (1962). On Karlowitz, Belgrade, and the survey commissions, see Fodor (2009); Ingrao (2000); Rothenberg (1966); Shaw & Shaw (1976).

veyors attended to trade corridors along the Danube, Sava, and Drava rivers to protect routes and customs revenues. These "high-stakes" segments are depicted in the second panel of Figure 4 and are less likely to satisfy the as-if-random assumption.

Yet these considerations were not uniform along the frontier. While many rivers were convenient lines of demarcation, not all were strategically or commercially salient. As Stoye observes, the Una was "too narrow a stream, with too many fords" (Stoye, 1994, p. 91), while the "boundary along the Sava and its Slavonian hinterland interested him less. [...] A world of flood, forest and marsh faced hillier but otherwise similar country across the river. In Slavonia there were no fortresses, but no need for them" (Stoye, 1994, p. 186). These marginal stretches, shaped primarily by geographic convenience rather than strategic calculation, constitute the most credible quasi-experimental settings. I label them "low-stakes" segments (Figure 4). Here, geography rather than politics plausibly determined imperial exposure, generating the local variation needed for causal inference about the long-run imprint of Ottoman versus Habsburg rule.

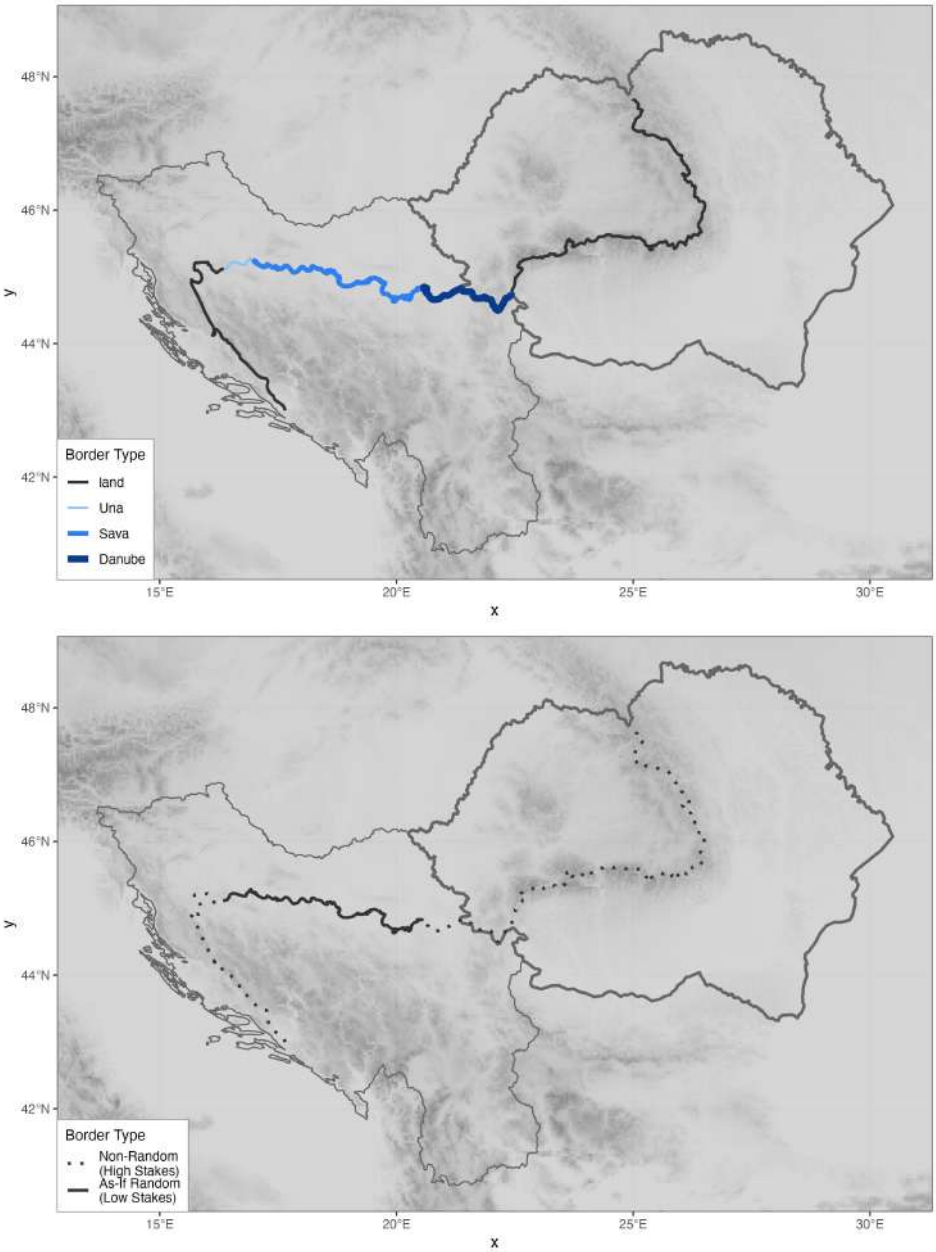
(3) Did negotiators have the capacity to shift the line toward or away from particular communities?

Negotiators had some room to maneuver, but were constrained by coalition politics, logistics, and the need for enforceable borders. Deliberations balanced symbolic parity with technical bargaining, and commissions implemented agreements by reconciling maps with terrain (Fodor, 2009; Stoye, 1994; Kurtaran, 2018). Where fortresses or trade routes were at stake, the line could be adjusted; elsewhere it tracked natural features to reduce future tensions.

Guided by these criteria, I pre-specify primary GRD estimates on the Una and Sava segments - surveyed, militarily marginal, and exhibiting smooth geographic covariates across the cutoff - while reporting adjacent segments as robustness checks.

This strategy privileges historical plausibility over blanket frontier coverage, strengthening the RD design. The segments most likely to meet the as-if-random assumption are presented in Figure 4.

Figure 4. *Border Types*



Covariate Balance at the Frontier

Identification requires continuity of potential outcomes and predetermined covariates in geographic location at the cutoff, and no precise manipulation of sorting around the boundary. I assess balance using five covariates that are fixed prior to treatment: elevation, slope, precipitation, temperature, and wheat suitability using the same RD2D procedure applied to literacy. Specifically, for each of the 164 evaluation points that I generate along the Habsburg-Ottoman frontier - one every 10 km (see the Estimation section as well), I estimate local discontinuities in these covariates using the same RD2D procedure applied to literacy outcomes.

Covariate balance varies sharply along the frontier. Differences are pronounced in mountainous and strategically salient areas - such as parts of the Carpathians and coastal Croatia - but negligible along the Una and Sava segments, where estimated discontinuities typically lie close to zero. Slope, precipitation, and temperature exhibit similar patterns. Wheat suitability displays greater variation overall but shows no systematic discontinuity along the Una-Sava corridor.

These patterns align with the historical analysis of border formation. Segments anchored in rivers for geographic convenience exhibit substantially smoother covariate profiles than segments shaped by strategic or economic considerations. In the results that follow, I explicitly distinguish estimates that satisfy covariate balance from those that do not. Specifically, the opaque points in Figure 5 denote estimates that satisfy four design-based credibility criteria: (1) idiographic historical support for conditional randomness, (2) balance on all five predetermined geographic covariates, (3) no evidence of sorting in density tests, and (4) sufficient effective sample support on both sides of the frontier. Transparent points indicate locations where one or more criteria fail. Statistical significance is reported but does

not determine design credibility.

Threats to Identification and Design Diagnostics

Several threats could undermine identification: endogenous border placement, selective sorting or migration near the cutoff, spatial spillovers, and misspecification arising from complex boundary geometry. I address each threat using a combination of historical evidence, design-based diagnostics, and robustness checks.

First, the frontier itself may be endogenous if it was drawn to capture systematically different communities. To reiterate, I do not assume as-if random assignment uniformly along the Habsburg–Ottoman border. Instead, I combine idiographic historical analysis of the border-making process with pre-specification of historically low-stakes segments—particularly the Una and Sava rivers - where demarcation plausibly reflected geographic convenience rather than strategic or economic selection. These segments anchor the causal interpretation; estimates from high-stakes segments are reported for transparency but should be interpreted cautiously.

Second, selective sorting around the frontier could generate discontinuities in population density or composition that would violate the regression discontinuity assumptions. I assess this threat using localized McCrary-style density tests computed at multiple evaluation points along the boundary rather than relying on a single global test. These tests indicate smooth population density at the cutoff for most segments (Appendix Section 6). In the main analysis, I therefore restrict causal interpretation to locations that satisfy both geographic covariate balance and density continuity. I also examine longer-run migration as a distinct threat to identification. The estimated Ottoman effects are robust to excluding districts with substantial population decline and to controlling for both absolute and relative population change (Appendix Section 7).

Third, geographic RD designs require that treatment effects are local to the boundary and not driven by discontinuous spillovers across it. In this setting, several features mitigate such concerns. The Habsburg–Ottoman frontier was a militarized imperial boundary that persisted for over a century (1699–1878), separating populations governed under distinct legal systems and bureaucratic structures. Even after 1878, portions of the frontier remained international borders between successor states (e.g., Romania–Serbia), while other segments became internal boundaries within successor polities. In both cases, institutional separation persisted for decades through distinct schooling systems and curricula that remained differentiated through the interwar period. I further focus estimation on segments anchored to natural barriers—particularly the Una and Sava rivers - where geography reinforced administrative separation. While residents could cross these rivers, doing so was costlier than movement within Habsburg or Ottoman territories. Any cross-border interaction that did occur—whether through trade, seasonal migration, or information exchange - would have operated gradually across space rather than generating step-function changes at the river's edge, thereby preserving the validity of local comparisons.

Any remaining spillovers would bias estimates toward zero. Diffusion of educational practices, teacher mobility, or information flows would tend to make adjacent communities more similar, attenuating rather than generating discontinuities. The RD2D estimates should therefore be interpreted as lower bounds on the true Ottoman effect. I evaluate these threats empirically using three diagnostics: localized density tests (Appendix Section 6), robustness checks excluding districts with substantial population change between 1931 and 1961 (Appendix Section 7), and tests for discontinuities in predetermined geographic covariates that would indicate endogenous boundary placement (Appendix Section 5).

Finally, recent work shows that distance-to-boundary GRDs can be biased when applied to irregular borders (Cattaneo *et al.*, 2025b,a). For this reason, the main analysis uses the bivariate location-score RD2D estimator, which is designed for boundary-based assignment and incorporates robust bias correction and data-driven bandwidth selection. Traditional distance-based RD estimates are reported as secondary checks and shown to exhibit substantial bias in many segments (see Appendix Sections 10 and 11). The RD2D estimator identifies local treatment effects under the assumption that potential outcomes are continuous in geographic location at the boundary and that treatment assignment for units near the cutoff does not affect outcomes for other nearby units (Cattaneo *et al.*, 2025a). The historical and geographic features described above support both requirements.

Estimation

I estimate local treatment effects using boundary discontinuity methods for geographic regression discontinuity designs following Cattaneo *et al.* (2025a). Specifically, I implement bias-corrected local polynomial estimators ($p = 1$) with robust bias correction based on local quadratic fits ($q = 2$), using triangular kernels and data-driven mean-squared-error–optimal bandwidth selection along the frontier. Inference relies on robust variance estimation allowing for within-county dependence, and I report both pointwise uncertainty and uniform confidence bands to account for simultaneous inference across multiple evaluation points along the frontier.

The RD2D framework treats the frontier as a series of local comparisons rather than imposing an arbitrary partition of the border into researcher-defined segments. A common approach in GRD applications is to aggregate the frontier into discrete segments and estimate pooled effects with segment fixed effects. While often con-

venient, such segmentations are frequently chosen for statistical tractability rather than to reflect historically meaningful variation. By contrast, RD2D allows treatment effects to vary flexibly along the boundary without imposing an arbitrary segmentation. I interpret this local heterogeneity using a pre-specified, historically grounded classification of segments into higher- and lower-stakes stretches, based on the process by which the boundary was negotiated and demarcated.

I generate 164 equally spaced evaluation points along the Habsburg-Ottoman frontier (one every 10 km). At each point, I estimate the discontinuity in literacy between localities on either side using the nonparametric bandwidth selected by the procedure, which defines the local comparison set (dashed neighborhoods in Figure 5). To account for spatial dependence induced by shared administration and measurement, I cluster the standard errors at the county level in Romania and at the appropriate administrative unit in Yugoslavia, matching the structure of the underlying census data.

Inference is based on the bias-corrected RD2D estimator; implementation details and additional specifications are reported in Appendix Section A.33. Recent theory cautions that distance-to-boundary GRDs can be biased near kinks and irregular geometries (Cattaneo *et al.*, 2025a). I therefore treat one-dimensional signed-distance specifications as secondary checks. Appendix Sections 10 and 11 report robustness to distance-based RD estimates (with and without kink adjustment), standard GRD implementations, and OLS/RD-robust alternatives.

Main Results

Figure 5 summarizes the design and main estimates. The first panel maps literacy in 1931 across Yugoslavia and Romania; darker shading north of the frontier indicates higher literacy in former Habsburg areas. The second panel overlays RD2D

estimates along the frontier; point shading reflects the estimated local Ottoman effect (more negative values indicate larger Ottoman disadvantages). Full covariate balance tests and density tests for all evaluation points are reported in Appendix Section 5 and 6.

Across the frontier, estimated discontinuities are predominantly negative, indicating lower literacy in former Ottoman areas in 1931. Magnitudes are especially large along Yugoslav segments of the border, where local gaps typically fall in the range of roughly negative 28-40 percentage points, whereas Romania exhibits smaller but still substantively meaningful discontinuities on the order of about negative 10 percentage points.[9] Figure 5 visualizes this spatial heterogeneity.

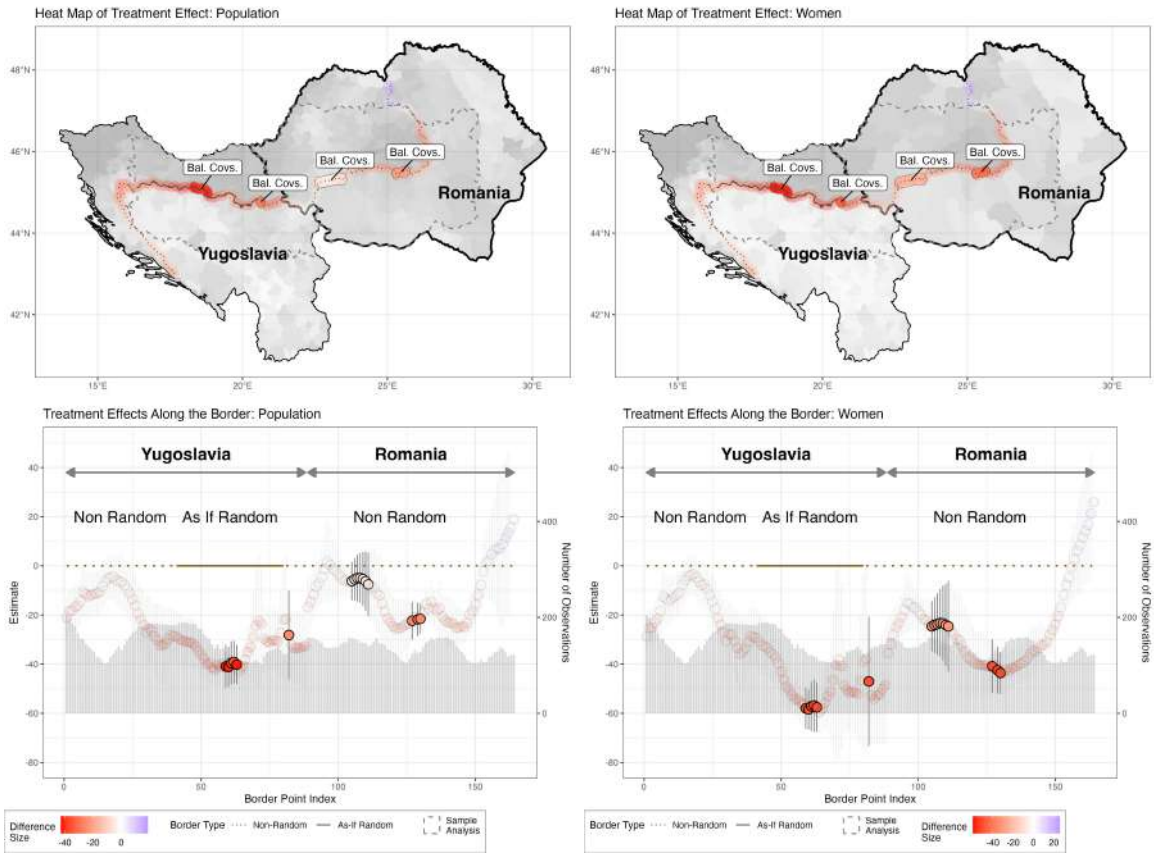
Gender disaggregation reveals even larger discontinuities. Along the Yugoslav land border, female literacy is approximately 35 percentage points lower on the former Ottoman side; in Romania, the corresponding gap is about 20 points. Among segments that plausibly satisfy as-if random assignment and pass the geographic balance tests, effects are especially pronounced. For example, along the Sava River in Yugoslavia the estimated female literacy gap is approximately 58 points, and along the Romanian frontier it reaches about 25 points.

These magnitudes reflect extremely low baseline female literacy in former Ottoman areas in 1931: roughly 15% in Yugoslavia and 38% in Romania, compared to 71% and 61% respectively on the Habsburg side.

How did the Ottoman–Habsburg literacy gap persist or attenuate between 1931 and 1991? Figure 6 and 7 trace literacy differences in Romania and Yugoslavia from the early twentieth century to the end of communism using both the non-parametric sample chosen with the GRD2D methodology (first map) and the full sample (second map). The figures report two sets of estimates: nonparametric RD2D estimates

[9]The full distribution of local estimates, bandwidths, and uncertainty intervals is reported in Appendix Section A.33.

Figure 5. *Romania and Yugoslavia Compared RD2D - Population and Women Literacy in 1931*



Notes: Points denote the same evaluation locations along the imperial border, displayed geographically in the top panels and indexed linearly in the bottom panels. Color indicates estimated treatment effects (red = negative, blue = positive), and transparency reflects local covariate imbalance. Background shading in the maps shows observed literacy rates for the total population and for women. Solid border segments correspond to low-stakes (as-if-random) sections, while dotted segments denote high-stakes sections. Dashed outlines mark illustrative local bandwidth neighborhoods used by the RD2D estimator. Grey bars in the coefficient plots indicate the number of observations contributing to each local estimate.

at border points with balanced covariates (black) and comparisons using the full treatment and control regions (grey).

In Romania, literacy differences were large in 1931: roughly −10 percentage points with the bivariate estimator and −14 with OLS. The gap narrowed steadily thereafter. By 1948, at the onset of communism, the difference was about −10 points, and by 1991 it had shrunk to −1 or −2 points. These patterns are visible in figure 6. The pattern is sharper in Yugoslavia as indicated in figure 7. Gaps of −38 to −43

points in the interwar period declined to about −19 percentage points by 1961, after a decade of communist reforms. By 1991, differences had narrowed further, to −3 to −7 points, as displayed in figure 7.

To quantify the convergence patterns shown descriptively in Figures 6 and 7, I estimate a difference-in-discontinuities model comparing how the Ottoman–Habsburg literacy gap evolved between 1931 and 1991 in each country. In levels, both countries closed a large share of the initial gap. Once mechanical ceiling effects are taken into account using log-odds transformations,[10] however, the contrast becomes stark. Romania closed 83 percent of its initial log-odds gap, whereas Yugoslavia closed only 27 percent. This divergence indicates that, despite similar centralized schooling efforts, the attenuation of Ottoman legacies was substantially more complete in Romania than in Yugoslavia, consistent with the conditional erosion framework.

Gender differences further illuminate this decline. The second columns in Figure 6 and 7 show the evolution of female literacy gaps. In Romania, the Ottoman–Habsburg difference was about −22 points in 1930 (using the GRD2D non-parametric sample), shrinking to −7 in 1977 and −3 in 1991. In Yugoslavia, the gap was far larger-around −59 points in 1930 - but diminished to roughly −10 by 1991. At near-universal literacy levels, a 5-point shortfall is not trivial: it means one in twenty adults remains without basic reading and writing who would otherwise be literate. The persistence of a gap of this size after four decades of socialist schooling campaigns is therefore substantively important.

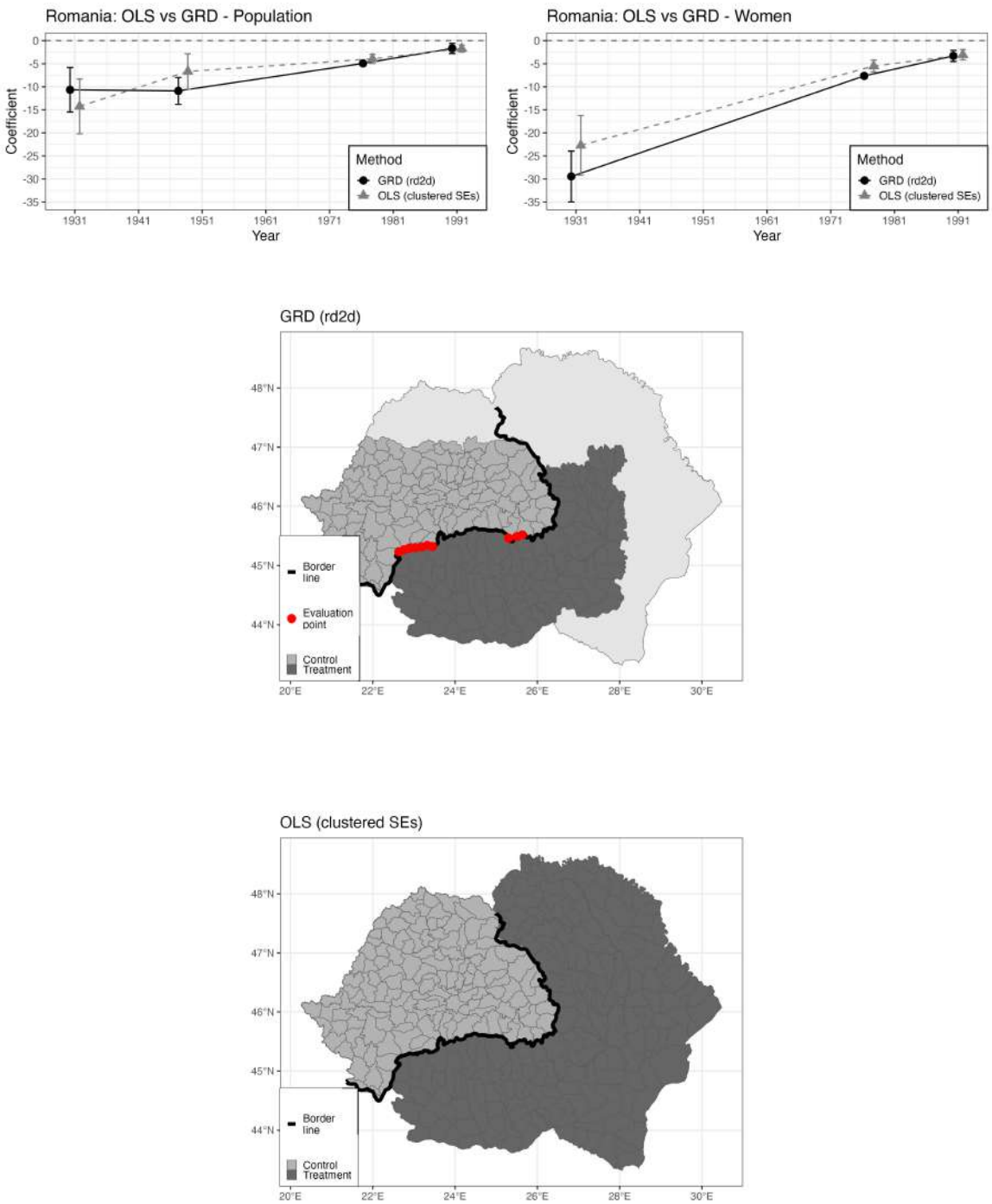
These patterns plausibly reflect targeted socialist campaigns to expand girls' education. Both Romania and Yugoslavia made schooling compulsory for girls and undertook rural initiatives to persuade peasant families to educate their daughters. In

[10] Log-odds ratios transform literacy rates to address the compression that occurs as outcomes approach 100 percent. The transformation is $\ln(p/(1-p))$, where p is the literacy rate. See Appendix Table A.2 for detailed estimates.

Romania, mass literacy drives and rural schooling were framed as central to socialist citizenship (Bucur, 2002). In Yugoslavia, women's emancipation was closely tied to education, with the Communist Party and the Antifascist Women's Front spearheading campaigns in rural areas (Brunnbauer, 2000; Poulton & Taji-Farouki, 2000). More broadly, communist modernization projects across both countries sought to reduce gender disparities and increase women's labor force participation (Lampland, 1995; Verdery, 1995).

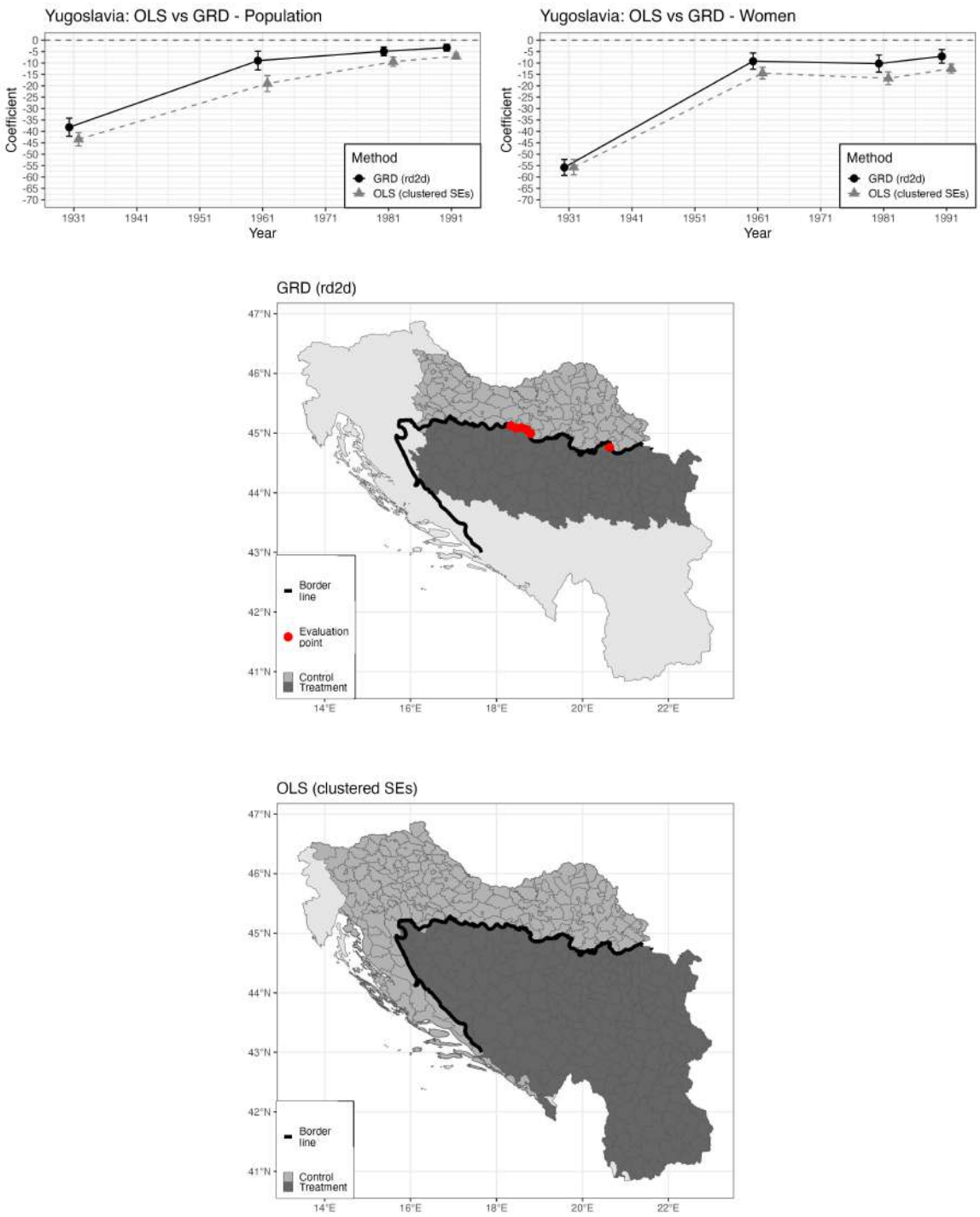
These trajectories are consistent with conditional erosion: centralized socialist schooling reduced the Ottoman literacy disadvantage in both settings, but attenuation was markedly more complete in Romania than in Yugoslavia once ceiling effects are accounted for.

Figure 6. *Evolution of Literacy Gaps in Romania: Population vs. Women*



Notes: Panels show estimated effects of Ottoman-rule exposure on literacy. The line plots report coefficients for Romania over time for the total population (left) and women (right). Black series: RD2D estimates of the local average treatment effect at boundary evaluation points that satisfy covariate-balance and McCrary density test criteria; gray series: OLS estimates with county-clustered standard errors. Vertical bars are 95% confidence intervals. The lower-left map displays the RD2D analysis sample and boundary evaluation points; the lower-right map displays the full sample used for the OLS specifications.

Figure 7. *Evolution of Literacy Gaps in Yugoslavia: Population vs. Women*



Notes: Panels show estimated effects of Ottoman-rule exposure on literacy. The line plots report coefficients for Yugsoalvia over time for the total population (left) and women (right). Black series: RD2D estimates of the local average treatment effect at boundary evaluation points that satisfy covariate-balance and McCrary density test criteria; gray series: OLS estimates with county-clustered standard errors. Vertical bars are 95% confidence intervals. The lower-left map displays the RD2D analysis sample and boundary evaluation points; the lower-right map displays the full sample used for the OLS specifications.

5 Mechanisms

This section evaluates the empirical plausibility of the mechanisms proposed in the theory. It does not seek to identify causal mechanisms in a formal sense. The channels under consideration - administrative capacity, state–society alignment, and demand-side compliance - are not experimentally manipulated, and the analysis does not estimate mediation effects. Instead, the goal is to assess whether patterns of persistence and attenuation in literacy align with these mechanisms and are difficult to reconcile with leading alternatives.

The analysis proceeds in three steps. First, I document inherited differences in administrative capacity, showing that bureaucratic density was substantially lower on the former Ottoman side in 1931, consistent with a legacy of weaker state penetration that plausibly constrained early human capital formation. Second, I examine variation in state–society alignment. Focusing on Yugoslavia, I show that post-imperial convergence was slowest in districts with larger Muslim populations, consistent with persistent misalignment following status reversal under socialist rule. Third, I distinguish between supply - and demand-side explanations using microdata, assessing whether the persistence of educational gaps reflects continued limitations in state provision or lower compliance rooted in historical mistrust of state institutions.

5.1 Administrative Capacity

I begin with the administrative capacity channel. Figure 8 documents lower bureaucratic presence on the Ottoman side (Yugoslavia: approx. −3.7 p.p. gap relative to a 2.1% mean; Romania: approx. −0.53 p.p. relative to a 0.66% mean). This pattern is consistent with weaker administrative penetration under the late Ottoman system,

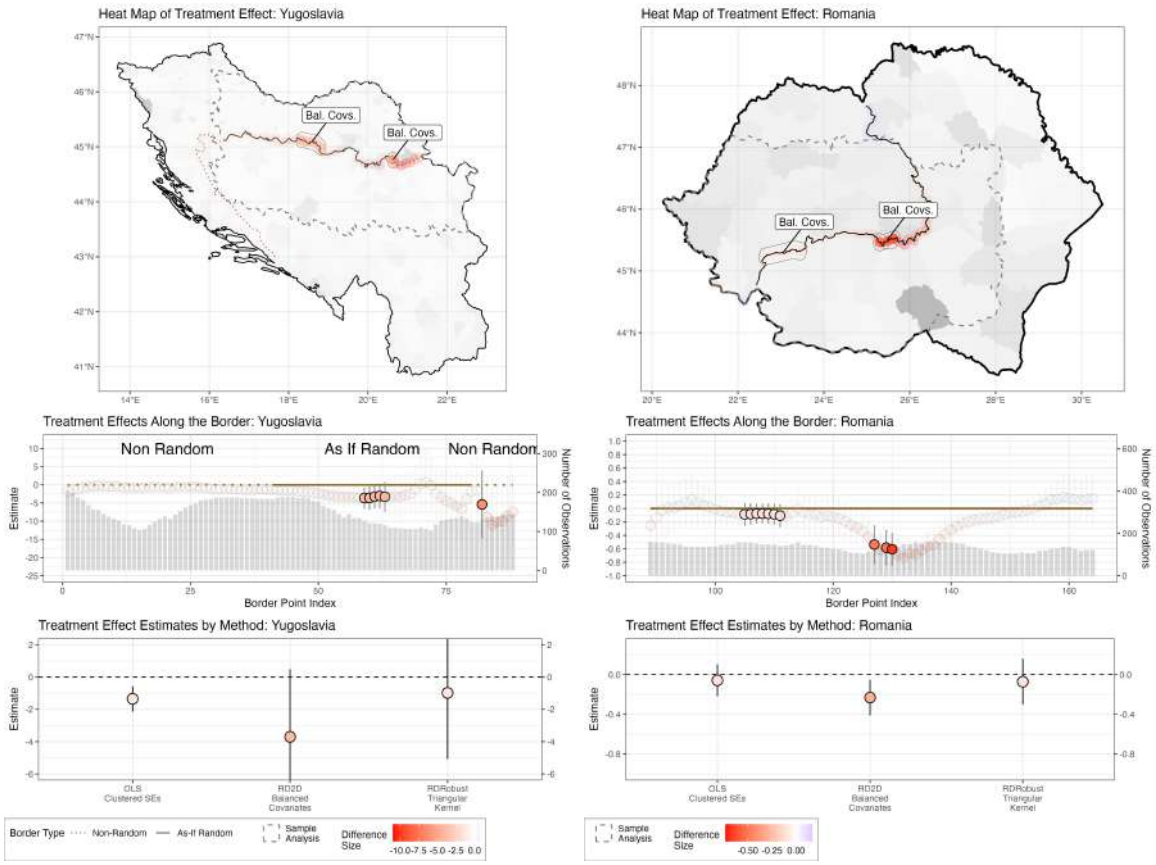
where reliance on local intermediaries fragmented governance and limited bureaucratic reach beyond core centers (Findley, 1996; İnalcık, 1996; Pamuk, 2001). The lower bureaucratic density is fully compatible with slower gains in literacy on the Ottoman side. As discussed above, the Ottoman Empire bequeathed both weaker administrative infrastructure and distinctive social structures through the millet system - legacies that operated jointly. Accordingly, bureaucratic density should be understood not as an independent determinant of literacy, but as a contextual constraint whose consequences depended on local alignment.

Both literacy and bureaucratic density are measured in 1931, raising the possibility that low literacy might itself have limited demand for bureaucrats rather than the other way around. The mutual reinforcement cannot be fully ruled out. However, the historical sequence strongly suggests that bureaucratic weakness preceded low literacy.

On the Ottoman side, provincial governance in the Balkans rested on patrimonial rule and fiscal extraction: from *timar* to *iltizam*, and - after 1695 - *malikâne* or lifetime farms that empowered local power holders and limited salaried administration (İnalcık, 1973; Barkey, 1996; Darling, 1996). Education remained largely a confessional enterprise funded through religious endowments, with state schools expanding only after the mid-nineteenth century and remaining thin in rural Balkan provinces before 1912 (Findley, 1980; Fortna, 2016; Somel, 1997).

By contrast, Habsburg cameral reforms under Maria Theresa and Joseph II built a salaried, examination-based civil service; district offices (*Bezirkshauptmannschaften*) and standardized fiscal and population records spread from the mid-eighteenth century (Ingrao, 2000; Judson, 2016). Compulsory primary schooling under the *Allgemeine Schulordnung* (1774) and *Ratio Educationis* (1777; rev. 1806) integrated education into a centralized bureaucracy well before 1931 (Deak, 2014).

Figure 8. *State Capacity in Yugoslavia and Romania, 1931*



Notes: Points denote boundary evaluation locations used by the RD2D estimator, shown geographically in the top panel and ordered along the border in the middle panel. Point color indicates estimated local treatment effects (red = negative, blue = positive), and transparency reflects whether local balance conditions are satisfied and the McCrary density test does not reject continuity. Background shading reports observed percentage of people employed in public administration (Romania) and percentage public servants (Yugoslavia). Solid border segments denote low-stakes (as-if-random) sections and dotted segments denote high-stakes sections. Error bars represent uniform confidence bands, and shaded bars indicate the number of observations contributing to each local estimate. The bottom panel compares average treatment effects from RDrobust, OLS with county-clustered standard errors, and the RD2D estimator averaged across balanced boundary points (95% confidence intervals shown)

Thus, while the 1931 measures are cross-sectional, the historical record supports interpreting weaker bureaucratic density as a legacy of imperial administration that constrained subsequent educational investment, rather than the reverse.

5.2 Why Do Legacies Attenuate Less in Yugoslavia? Social Heterogeneity and Alignment

I now examine variation *within* Yugoslavia to assess whether the proposed mechanisms - state–society alignment and coordination costs - can account for differences in convergence. This within-country analysis provides a final probe of the alignment mechanism by exploiting subnational variation in social structure while holding constant national institutions, schooling policy, and overall regime capacity.

If the mechanisms are operative, literacy gaps should persist longest in districts with larger Muslim populations, indicating lower alignment with the Christian-dominated state, and in districts with higher religious fractionalization, reflecting greater coordination costs.

Figure 9 plots the treatment-by-religion dynamics in Yugoslavia for Muslims, Orthodox Christians using the non-parametric sample determined by the GRD2D. Muslims and Orthodox Christians are the two dominant religious groups on the Ottoman side of the frontier. The share of Catholics is comparatively small (6.7 percent on the Ottoman side in 1931), which is why I do not report the corresponding interaction. Each panel in Figure 9 presents two specifications: one without controls (shown in black), and one including elevation and population density (shown in grey). Standard errors are clustered at the county level in both cases.

The first panel reports the relationship between the share of Muslims and literacy from 1931 to 1991.[11] Literacy levels among Muslims are initially very low in 1931,[12] but the gap narrows steadily over time as communist adult-education initia-

[11]The independent variable in the first panel is the percentage of Muslims, not the interaction between the Ottoman-territory indicator and the Muslim share, since there are virtually no Muslims in the former Habsburg territory.

[12]A one–percentage-point increase in the Muslim share of the population is associated with a 0.85 percentage-point decrease in the literacy rate.

tives and universal-schooling campaigns expanded (Donia & Fine, 1994; Velikonja, 2003; Lampe, 1996).

The second panel displays the interaction between the Orthodox share and Ottoman treatment. Orthodox communities exhibit much higher literacy levels in the 1930s than non-Orthodox communities[13], but these differences also diminish over time.

The negative association between Muslim population share and literacy therefore points to the presence of a potentially institutionally misaligned group, rather than to diversity effects per se. A prominent critique of the "diversity effects" literature argues that fractionalization indices often proxy for the presence of such disadvantaged groups rather than capturing heterogeneity as an independent mechanism (Kustov & Pardelli, 2018). To adjudicate between these interpretations, I also estimate models that include religious fractionalization and group population shares separately. Consistent with the alignment mechanism, the Muslim population share emerges as the most informative predictor, while fractionalization plays a limited role.

The third panel in Figure 9 plots the interaction between religious fractionalization and Ottoman treatment. In 1931, higher religious fractionalization is strongly and negatively associated with literacy in former Ottoman districts, indicating that heterogeneity initially compounded educational disadvantage under Ottoman institutional legacies. This negative association persists, though at a substantially reduced magnitude, into the postwar period. By the 1980s and 1990s, however, the interaction attenuates to zero and is no longer statistically distinguishable from no effect. This temporal pattern suggests that religious fractionalization per se was not a persistent barrier to educational expansion. Instead, its early negative associ-

[13]In former Ottoman areas, a one percentage-point increase in Orthodox share is associated with a 0.21 percentage-point increase in literacy.

ation reflects the presence of specific misaligned minority groups within Ottoman territories-an effect that diminished as mass education expanded and institutional constraints weakened over time.

5.3 Supply vs. Demand of Literacy

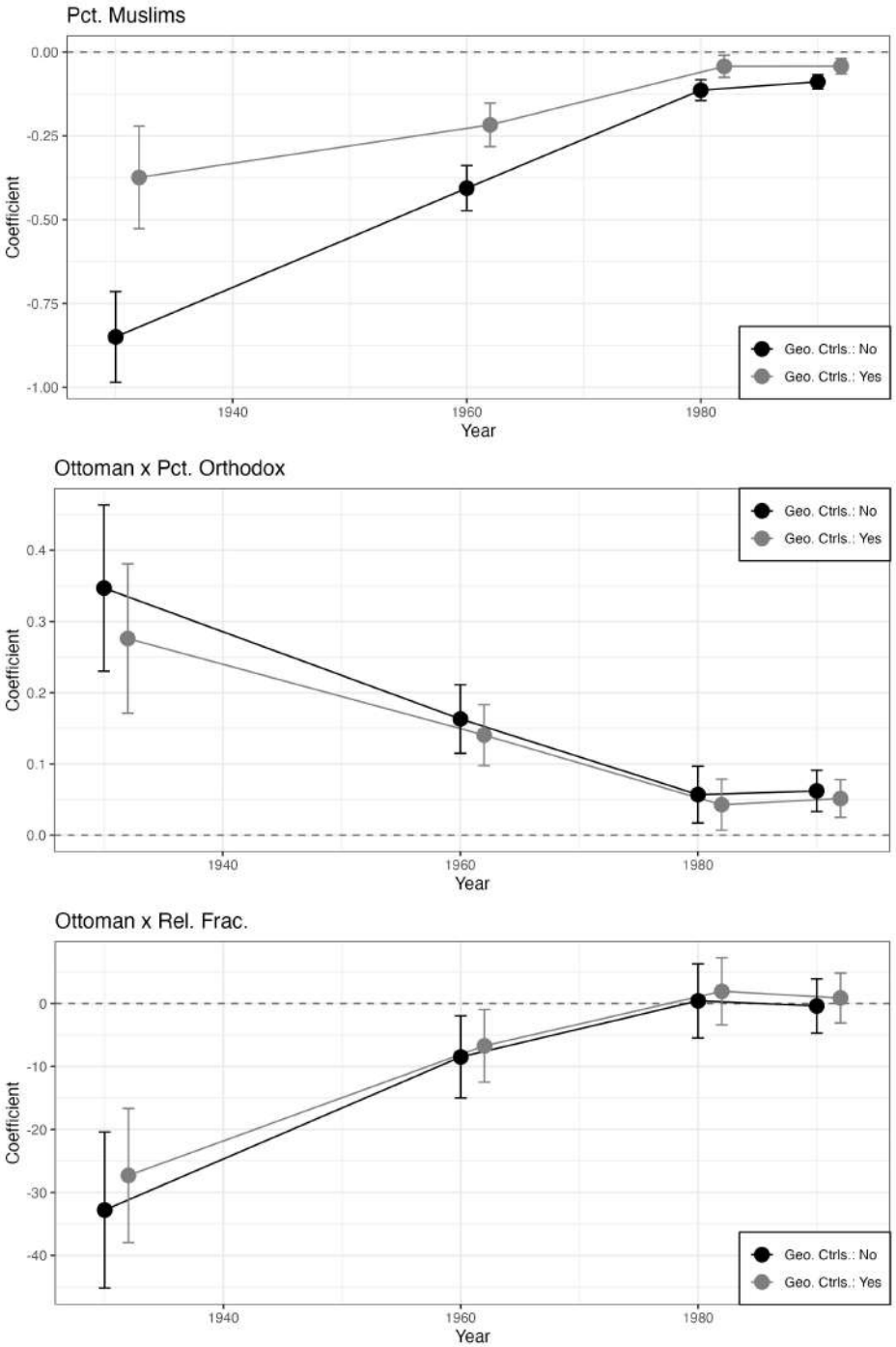
An important distinction is whether the observed literacy gap - particularly the slower convergence among Muslim communities - reflects supply-side constraints, such as limited school availability, or demand-side frictions, including lower compliance or mistrust of state institutions. I begin by assessing the supply-side channel using district-level data on school provision in 1961 and 1981.

Regressions of schools per capita on the Muslim population share reveal persistent disparities in physical school infrastructure. Even after controlling for geography, districts with larger Muslim populations had fewer schools per capita in both periods, indicating that infrastructure gaps narrowed only slowly over time (Figure 10). By contrast, regressions using teachers per capita show a pronounced negative association with Muslim population share in 1961 that disappears by 1981, suggesting that staffing levels converged more rapidly than physical school density.

Taken together, disparities in physical school infrastructure narrowed over time but did not disappear entirely, while staffing levels converged more rapidly than school density. These supply-side patterns alone, however, are insufficient to account for the persistence of literacy and educational attainment gaps into the late socialist period.

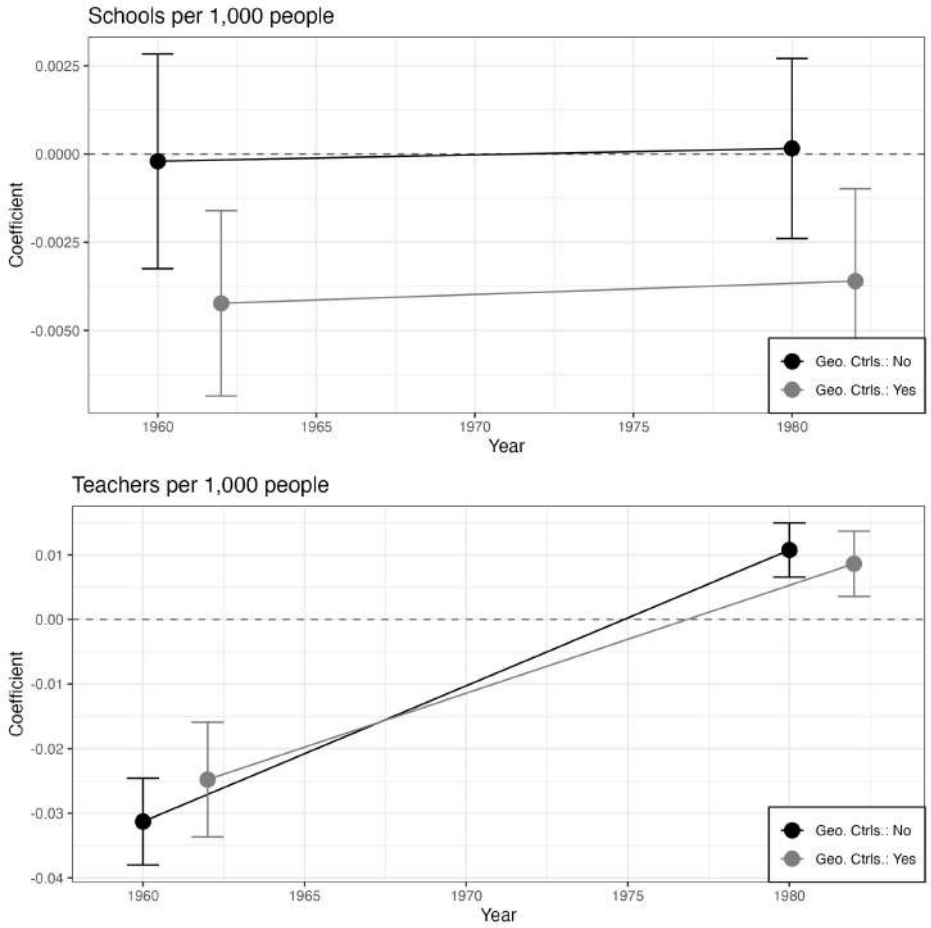
I next leverage individual-level survey microdata to assess whether the legacy gaps observed in 1991 are concentrated among cohorts educated under socialism. Because these data are measured long after the schooling campaigns themselves, they do not identify mechanisms directly; instead, they provide a cohort-based di-

Figure 9. *Heterogeneous Effects on Literacy by Religion in Yugoslavia*



Notes: Points report coefficient estimates from OLS regressions estimated on the analysis sample. Black series show specifications without geographic controls; gray series include controls for population density and elevation. Error bars denote 95% confidence intervals. The dependent variable is the literacy rate. The main regressor is the share of Muslims (first panel), the interaction between Ottoman exposure and the Orthodox share (second panel), and the interaction between Ottoman exposure and religious fractionalization (third panel)

Figure 10. *Effect of Muslim Concentration on Educational Infrastructure in Yugoslavia*

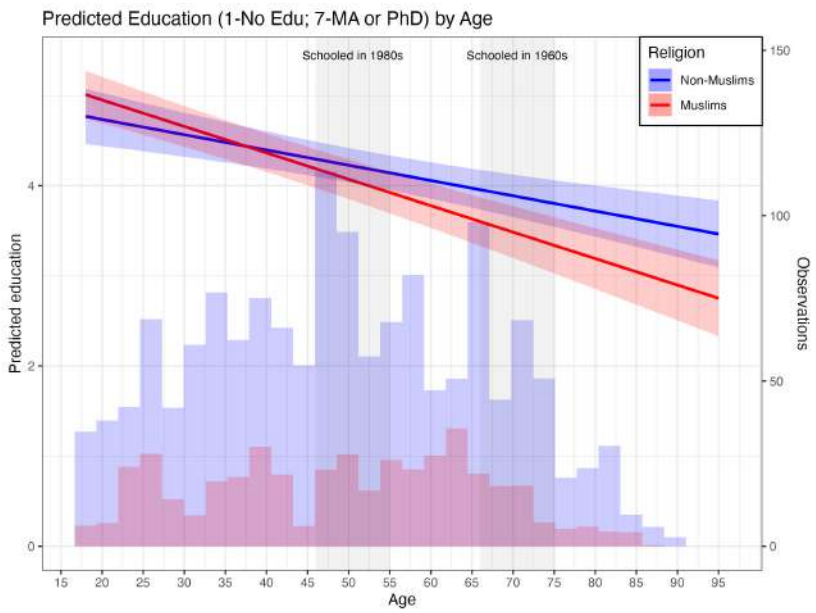


Notes: Points show OLS coefficients relating Muslim population share to schools (top) and teachers (bottom) per 1,000 inhabitants. Gray series include geographic controls; black series do not. Error bars indicate 95% confidence intervals.

agnostic that helps distinguish "supply-only" accounts from explanations that also imply persistent participation and compliance frictions. Figure 11 plots completed schooling by Muslim status across the age distribution. The evidence indicates that, despite large expansions in schooling provision, Muslim cohorts educated in the 1960s–1980s continue to report lower attainment than their non-Muslim counterparts, suggesting that convergence in inputs alone cannot account for the persistence of educational gaps.

This pattern raises the question of whether communities differed in their willingness to participate in the schools that did exist. The alignment mechanism implies

Figure 11. *Predicted Education by Age and Muslim Status*



Notes: Predicted education levels are obtained from a weighted OLS model with Muslim status interacted with age and controls for sex and income. Shaded areas indicate 95% confidence intervals; bars show the age distribution. Education is coded on an eight-point scale ranging from no education (1) to master's degree or PhD (8).

that minority communities whose identities were historically misaligned with the dominant state would have been less willing to enroll children and less trusting of state-run institutions, even as access to schooling expanded.

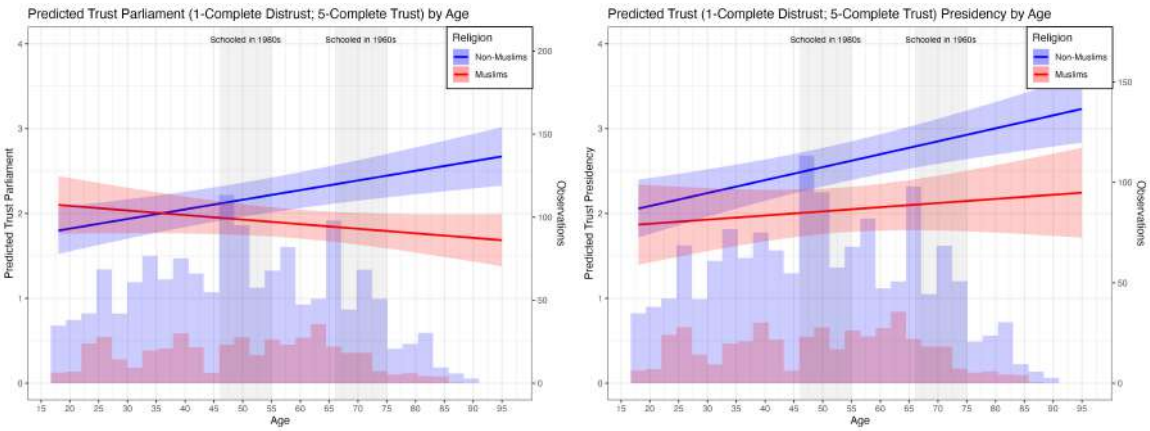
Qualitative historical evidence strongly supports this interpretation. In her detailed ethnography of a central Bosnian Muslim village, Bringa (1995) documents how older generations perceived state institutions-and particularly state schools-as culturally distant and morally threatening. As she notes, "To the teacher in state school a 'Muslim' was a member of a [...] communist and nonreligious group. To the mosque teacher [...] Muslim identity was primarily an Islamic one" (Bringa, 1995, p. 204). Moreover, villagers understood that "an individual could not at the same time hold communist party membership and be a practicing Muslim" (*ibid*). These accounts describe precisely the form of early postwar misalignment that would depress participation in state schooling among older Muslim cohorts which is also

compatible with the quantitative results in Figure 11.

Cohort patterns in institutional trust from the LITS surveys (2010/2016/2025) provide further evidence that alignment reflects historical legacies rather than purely communist-era discrimination. After controlling for gender, income, and survey-year fixed effects and clustering standard errors by PSU, older Muslims (age 55+, born before 1955 and educated during the 1960s–1970s) exhibit substantially lower trust in the presidency and parliament than older non-Muslims (Figure 12). These are precisely the cohorts whose educational attainment contributed most to the 1991 literacy gaps. Among younger adults (born after 1975 and educated primarily after communism), the pattern reverses: young Muslims report higher trust in these institutions than their non-Muslim counterparts.

This generational reversal is difficult to reconcile with accounts that attribute mistrust primarily to discriminatory policies during socialism. While older cohorts educated under communism do exhibit lower trust, this mistrust persists decades after the regime's collapse (measured 2010–2025), suggesting deeper historical roots rather than contemporaneous policy effects. At the same time, the reversal of the trust gap among younger cohorts—educated primarily after the end of socialism—indicates that misalignment was not fixed but gradually eroded as historical distance from early postwar status reversal increased and as post-Yugoslav institutions evolved. This pattern is consistent with inherited mistrust rooted in early postwar misalignment, when Muslim communities perceived state schools as vehicles of cultural domination, gradually weakening across cohorts as that historical experience receded.

Figure 12. *Predicted Trust in Parliament and the Presidency by Muslim Status*



Notes: Predicted trust are obtained from a weighted OLS model with Muslim status interacted with age and controls for sex and income. Shaded areas indicate 95% confidence intervals; bars show the age distribution. Trust is coded on a five-point scale ranging from complete distrust (1) to complete trust (5).

Alternative Explanations

I consider three alternative explanations that could account for why literacy converged more fully in Romania than in Yugoslavia, independent of our argument about state–minority alignment: (i) selective migration; (ii) institutional constraints arising from Yugoslavia's federal structure; and (iii) differences in regime coerciveness under communism. Each explanation yields clear empirical implications that I evaluate directly.

A first alternative explanation is selective migration. If selective migration were driving convergence, Ottoman-treated areas should experience systematically different population trajectories or compositional changes correlated with treatment. Across both cases, selective migration is unlikely to account for the estimated Ottoman disadvantage or for the Romania–Yugoslavia divergence in convergence. In Yugoslavia - where postwar mobility was most plausibly consequential - I explicitly reconstruct population change between 1931 and 1961 on a harmonized grid and show that the Ottoman effect is essentially unchanged when excluding areas with

population decline and when controlling for absolute and percentage population change as well as religious-group compositional shifts. In Romania, I directly test whether baseline population mobility is correlated with treatment using the 1930 census floating population *populația flotantă* measure and find no statistically significant association between Ottoman exposure and floating population. The results are presented in section 7 of the appendix.

A second alternative explanation is that Yugoslavia's federal structure weakened the center's capacity to equalize human capital across regions. A federalism-based account predicts uniformly slower convergence across Yugoslavia relative to unitary Romania, and—critically - similar convergence trajectories across Yugoslav republics operating under the same constitutional and fiscal framework. This prediction is not supported by the data. First, Yugoslav federalism became more decentralized only after the 1960s, whereas most literacy convergence occurred earlier, during the initial decades of socialist mass schooling. Second, and more decisively, within Yugoslavia, Serbia - governed under the same federal constitution and subject to the same decentralization constraints and education policies as Bosnia and Macedonia - exhibits a convergence trajectory that closely mirrors Romania's and diverges sharply from Yugoslavia as a whole. Unlike Bosnia and Macedonia, the former Ottoman regions of Serbia were comparatively homogeneous and predominantly Orthodox, and therefore experienced much less the post-imperial status reversal of a large Muslim population. Because federalism is constant across Serbian, Bosnian, and Macedonian districts while ethnoreligious composition varies substantially, this within-federation comparison directly falsifies a federalism-based explanation. The evidence instead supports the argument that convergence depends on state–minority alignment rather than regime structure per se. Full results are reported in Appendix Section 8.

A third alternative explanation is that Romania achieved more complete convergence because its communist regime was substantially more repressive than Yugoslavia's, enabling more effective implementation of mass literacy campaigns through coercion rather than consent. Because regime coerciveness varies primarily at the national level, however, this explanation cannot account for systematic within-country differences in convergence. If regime coerciveness were the key driver of convergence, repression should be a necessary condition for rapid literacy gains. This explanation likewise finds little empirical support. First, holding regime repression constant within Yugoslavia, literacy converged rapidly in socially homogeneous Serbian districts characterized by high state–society alignment, despite operating under Tito's comparatively less coercive regime. This indicates that Romania-level repression is not a necessary condition for convergence. Second, a placebo test using individual-level LITS data from Romania shows no evidence that former Ottoman areas were more likely than Habsburg areas to experience political persecution, imprisonment, or labor camp exposure under communism. This rules out the possibility that convergence in Romania was driven by targeted coercion in Ottoman regions. Full results are presented in Appendix Section 9.

Taken together, these tests rule out migration, federal structure, and regime coerciveness as primary drivers of differential convergence, leaving state–minority alignment as the explanation most consistent with the full set of empirical patterns.

6 Discussion

Imperial legacies do not erode uniformly under centralized successor regimes. This article advances a conditional theory of post-imperial convergence, showing that inherited administrative capacity is necessary but insufficient to equalize outcomes

unless successor states govern populations that perceive state institutions as legitimate and aligned. Using the Habsburg–Ottoman frontier as a comparative setting, I show that early twentieth-century literacy gaps reached nearly 40 percentage points and closely tracked imperial exposure. These findings resolve a long-standing puzzle in the literature on imperial legacies: why deeply entrenched educational inequalities disappear rapidly in some post-imperial contexts, yet persist for decades in others despite comparable investments and ideological commitment. The evidence indicates that convergence depends on the interaction between administrative capacity and state–society alignment, rather than on capacity alone.

These gaps reflected uneven administrative capacity: thinner Ottoman bureaucracies constrained school provision, while denser Habsburg infrastructures supported earlier standardized schooling. Under communism, literacy gaps narrowed on both sides of the frontier, but convergence was uneven. It was nearly complete in relatively homogeneous Romania, yet remained substantial in socially diverse Yugoslavia, where misalignment between Muslim minorities and the Christian-identified state limited educational expansion despite comparable regime capacity. I argue that this pattern can be explained by the interaction between inherited administrative weakness and state–society alignment. Statistical analysis confirms that Ottoman territories had lower bureaucratic density in 1931, and that within Yugoslavia, literacy gaps persisted longest in districts with larger Muslim populations—precisely where status reversal had transformed an imperial ruling group into post-imperial minorities. Survey evidence supports this interpretation, showing that older Muslim cohorts report lower trust in state institutions, though this pattern reverses among younger generations educated after communism ended.

The core theoretical contribution of this article is to distinguish administrative capacity from state–society alignment. Administrative capacity determines

what states can supply—schools, teachers, bureaucratic oversight, and standardized curricula—while alignment determines whether those supplies translate into realized outcomes. Where populations perceive state institutions as legitimate, representative, or culturally congruent, centralized investments are more likely to generate compliance and convergence. Where alignment is weak, comparable investments yield diminishing returns, even under regimes with strong formal capacity.

I emphasize the implications of alignment for educational convergence under authoritarian rule, but the argument draws on equilibrium dynamics. In a longer time frame, misalignment between minorities and the state is both a cause of uneven outcomes and an effect. When ethnic or religious minorities mistrust state institutions, officials cannot easily verify their needs or ensure that invested resources will generate compliance, which lowers incentives to target these communities. The resulting uneven provision further reduces minorities' willingness to participate in state programs. The theory is most applicable to contexts where centralized regimes inherit weak administrative infrastructures and govern populations divided along ethnoreligious lines—conditions that characterized much of post-imperial state-building.

The argument therefore has implications beyond the Habsburg–Ottoman borderlands, but only under clearly specified conditions. It applies most directly to post-imperial and post-authoritarian settings in which centralized regimes inherit uneven administrative infrastructures and govern populations divided along historically entrenched ethnoreligious lines. In such contexts, formal administrative capacity determines what states can supply, but state–society alignment conditions whether those supplies translate into compliance and convergence. Where state institutions are perceived as legitimate, representative, or culturally congruent, centralized investments can generate rapid equalization. Where alignment is weak, comparable investments yield diminishing returns, even under regimes with strong

formal capacity. Analyses of post-imperial development that abstract from alignment therefore risk overstating the equalizing power of centralized state-building alone.

These scope conditions help clarify why ambitious educational interventions often fail to erase historical disadvantage in contemporary cases such as Afghanistan, Iraq, northern Nigeria, or Syria. In such settings, investments in schools, teachers, and infrastructure are unlikely to produce convergence unless communities perceive education as serving their interests rather than as instruments of cultural assimilation or political control. Mexico's prolonged efforts to extend Spanish-medium schooling into Indigenous regions illustrate the same dynamic: despite repeated twentieth-century campaigns and substantial federal investment, literacy and educational participation improved slowly in Chiapas, Oaxaca, Guerrero, and the rural highlands, where centralized schooling was widely perceived as externally imposed and culturally intrusive. The persistence of inequality in these cases reflects not insufficient state effort, but the limits of technocratic expansion in the absence of state–society alignment.

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Appendix

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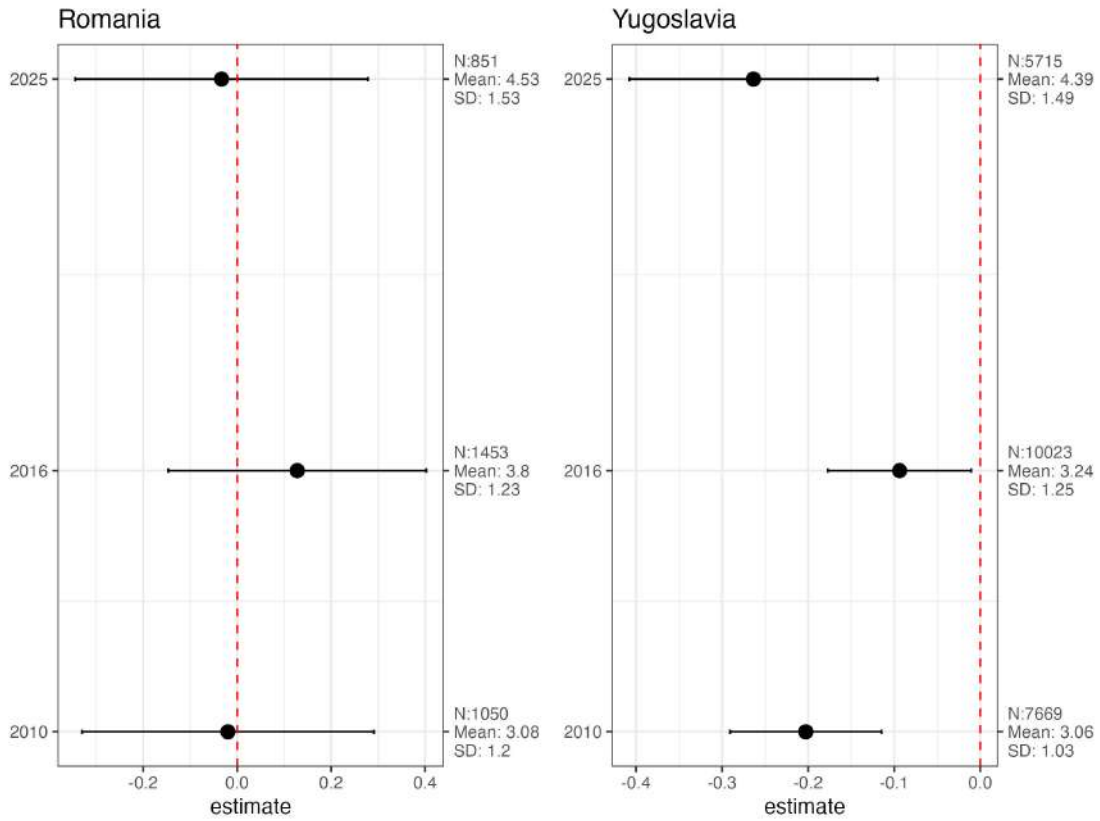
1 The Puzzle Revisited

Figure 1 in the main text documented substantial regional variation in literacy across Eastern Europe. In some successor states, such as Romania, the gaps evident in 1931 had largely disappeared by the end of the communist period. In others, such as Yugoslavia, they persisted. While Figure A.13 reports estimates based on census data, analogous patterns emerge in the LITS microdata (2010, 2016, 2025).

The dependent variable in these regressions is the highest level of education completed, coded in the following way: 1 - No degree / No education; 2 - Primary education; 3 - Secondary education; 4 - Post-secondary non-tertiary education; 5 - Tertiary education (not a university diploma); 7 - Bachelor's degree or more; and 8 - Master's degree or PhD. In Romania, the estimated coefficients of Ottoman rule are small in magnitude and statistically indistinguishable from zero across all survey waves (2010: −0.020, $p = 0.90$; 2016: 0.128, $p = 0.36$; 2025: −0.033, $p = 0.84$), indicating the absence of a durable Ottoman effect. By contrast, in Yugoslavia the estimated coefficients are consistently negative and statistically significant (2010: −0.203, $p < 0.001$; 2016: −0.094, $p = 0.03$; 2025: −0.264, $p < 0.001$). Substantively, these values correspond to a reduction of approximately 0.1–0.3 of an educational level, even after controlling for observable covariates.

Figure A.13. *Borders*



Robustness: Log-Odds Transformation

A potential concern with the convergence analysis is that percentage-point gaps mechanically compress as literacy approaches universal levels-one cannot have more than 100% literacy. To address this, I re-estimate all models using log-odds transformations of literacy rates. The log-odds transformation is $\text{logit}(p) = \ln(p/(1-p))$, where p is the literacy rate (expressed as a proportion between 0 and 1). To avoid undefined values at the boundaries, I apply a small adjustment: $p_{\text{adj}} = \max(\min(p, 0.9999), 0.0001)$ before computing the log-odds.[14] This transformation maps the bounded [0,1] interval onto the real line and is the standard econometric approach for modeling proportions near ceiling or floor effects.

I then re-estimate all four regression discontinuity models-Romania 1931, Romania 1991, Yugoslavia 1931, and Yugoslavia 1991-using log-odds literacy as the dependent variable instead of percentage-point literacy rates. Standard errors remain clustered at the county level (Romania) and *srez* level (Yugoslavia 1991), matching the specification of the main models.

Table A.2 presents the results. The log-odds specification reveals that convergence was far more complete in Romania than suggested by raw percentage points alone. While Romania closed 88% of its percentage-point gap and Yugoslavia closed 84%-a difference of only 4 percentage points-the log-odds analysis shows Romania closed 83% of its gap versus only 27% in Yugoslavia, a difference of 56 percentage points. This dramatic divergence indicates that Yugoslavia's apparent convergence in percentage-point terms substantially reflects mechanical compression rather than genuine equalization. Even at high literacy levels, Yugoslavia retained a large residual Ottoman penalty (log-odds gap of 1.59, equivalent to approximately 8 percentage points when evaluated at 90% baseline literacy),[15] whereas Romania's residual gap was near zero (log-odds gap of 0.11, equivalent to approximately 1 percentage point at 90% baseline literacy).

These findings confirm that convergence was meaningfully and substantially more complete in Romania than in Yugoslavia, supporting the paper's core theoretical claim that ethnoreligious homogeneity and state alignment facilitate the erosion of imperial legacies. The log-odds transformation demonstrates that this conclusion is robust to accounting for the mechanical compression of percentage-point differences at high literacy levels.

[14]This adjustment prevents numerical issues when literacy rates approach exactly 0% or 100%, while having negligible impact on the substantive results since observed rates in the data are well within these bounds.

[15]To interpret the log-odds gap: at a baseline literacy rate of 90% (log-odds = 2.20), a log-odds difference of 1.59 evaluated at 90% baseline implies a 7.8pp gap yields a log-odds of 3.79, which corresponds to approximately 97.8% literacy-a gap of 7.8 percentage points.

Table A.2. *Convergence in Ottoman-Habsburg Literacy Gaps: Percentage Points vs. Log-Odds*

	Percentage Points			Log-Odds		
Country	1931 Gap	1991 Gap	% Closed	1931 Gap	1991 Gap	% Closed
Romania	14.3***	1.7***	87.9%	0.65***	0.11***	82.7%
	(2.97)	(0.39)		(0.14)	(0.03)	
Yugoslavia	43.5***	7.0***	83.9%	2.17***	1.59***	26.9%
	(1.48)	(0.64)		(0.09)	(0.19)	
Difference			4.0pp			55.8pp

Notes: Standard errors in parentheses, clustered at county level (Romania) and *srez* level (Yugoslavia). The "Gap" columns show the estimated Ottoman disadvantage (positive values indicate lower literacy on the Ottoman side). Log-odds transformation is $\ln(p/(1-p))$ where p is the literacy rate. "% Closed" indicates the percentage of the initial gap that was eliminated by 1991. *** p<0.01, ** p<0.05, * p<0.10.

2 Data Collection

To measure the duration of Ottoman rule, I developed a Geographic Information Systems (GIS) dataset illustrating the evolving borders of the Ottoman Empire from its inception in 1326 under Osman I to its dissolution in 1922. This dataset was meticulously compiled through the scanning and digitizing 152 maps sourced from repositories such as the Ottoman Archives and the University of Chicago Map Collection. Some of these were individual maps, while others were part of historical atlases. Similarly, some offer an overall depiction of the empire's extent, while others are detailed representations of Ottoman rule within individual countries. The languages of these atlases include English, German, French, Serbo-Croatian, Hungarian, Italian, Turkish, Ottoman Turkish, Latin, etc. These maps offer a granular perspective, delineating regions under Ottoman control with remarkable precision. I georeferenced and compared different maps against each other, which is why this data is more accurate than other measures capturing the spread of empires at the turn of centuries (e.g., Nüssli (2002)). This approach is also much more fine-grained than country averages proposed by Dimitrova-Grajzl (2007) or the region averages employed by Grosjean (2011) based on Nüssli (2002), which only reveal border changes at the turn of the century.

Different maps were compared against each other, which is why this data is probably more accurate than other measures capturing the spread of empires at the turn of centuries (e.g. Nüssli (2002)).

The historical maps that were used for the creation of the map in figure 1 cover the following years:

- 14th Century: 1326, 1331, 1352, 1354, 1355, 1360, 1361, 1362, 1363, 1367, 1371, 1372, 1377, 1381, 1382, 1383, 1385, 1386, 1387, 1389, 1391, 1392, 1393, 1394, 1395, 1396, 1397, 1398
- 15th Century: 1402, 1415, 1416, 1417, 1428, 1430, 1439, 1442, 1443, 1444, 1448, 1450, 1453, 1455, 1456, 1458, 1459, 1460, 1461, 1462, 1463, 1466, 1469, 1470, 1471, 1475, 1476, 1478, 1479, 1480, 1481, 1482, 1485, 1499
- 16th Century: 1500, 1503, 1512, 1520, 1521, 1522, 1526, 1528, 1534, 1537, 1538, 1540, 1541, 1544, 1550, 1552, 1557, 1566, 1570, 1577, 1579, 1582, 1591, 1593
- 17th Century: 1606, 1617, 1646, 1648, 1664, 1669, 1672, 1683, 1684, 1685, 1686, 1687, 1688, 1689, 1690, 1691, 1692, 1693, 1695, 1697, 1699
- 18th Century: 1715, 1718, 1721, 1723, 1732, 1733, 1734, 1736, 1739, 1772, 1774, 1775, 1779, 1783, 1791, 1792, 1793, 1795, 1797
- 19th Century: 1800, 1803, 1805, 1807, 1809, 1810, 1814, 1820, 1829, 1830, 1834, 1838, 1855, 1856, 1860, 1863, 1864, 1878, 1880, 1881, 1882, 1885, 1889, 1895
- 20th Century: 1908, 1912, 1913, 1918, 1920.

Sources: American Geographical Society of New York (191-), Balta (1989, 1992); Barnes (2015); Bayerle (1973); Béranger (1990); Birken (1979, 1981), Great Britain. War Office. General Staff. Geographical Section. (1918); Cvijć (1918); Davies (1879), Delilbaşı & Arıkan (2001); Domanovszky (1940); Géza & Pál (2000); Géza (1999), Gopčević (1889), Halasi-Kun (1964), İnalcık *et al.* (2013), Johnston (1879, 189-), Káldy-Nagy & Mikesy (1960); Káldy-Nagy (1961); Káldy-Nagy (1971); Kiel (1993a,b, 1989), Kiepert (185-, 1869, 1876, 1880, 1881, 1895), İnalcık (1954); Lajos (1943, 1996); Langhans (1910); Laut (1991), Levasseur (1864); Liechtenstern & Amon (1795), Ljušić & Rajić (2003); Ljušić *et al.* (2006); Magocsi (2002), Unknown Author (1890), (Pitcher, 1972; Regan & Kaniški, 2003), Andrees (1912), Sarınay *et al.* (2001, 2003, 2006, 2007, 2009a,b); Spaho & Aličić (2007); Sršan & Hafizović (2001); Stefănescu *et al.* (1996a,b); Stojaković (1987), United States. Central Intelligence Agency (1908); Ward *et al.* (1912).

3 Methodology for Estimating Regional Male and Female Literacy Rates in Yugoslavia (1981-1991)

For Serbia and Kosovo in 1981 and 1991, I used female and male literacy rates from Republički Zavod Za Statistiku (1981, 1991). For Slovenia in 1981 and 1991, I used data from Statistični urad Republike Slovenije. [Statistical Office of the Republic of Slovenia] (1981, 1991). Finally, for Bosnia, I used female and male literacy figures from Federalni zavod za statistiku Federacije Bosne i Hercegovine (1998), which cover both 1981 and 1991. For Croatia, Macedonia, and Montenegro, I imputed male and female literacy rates using district-level data and national benchmarks, as detailed below.

Croatia (1981 and 1991): I estimated sex-specific literacy rates from district-level data in Savezni Zavod Za Statistiku (1987) and Croatian Bureau of Statistics (2005). Because district tables did not report male and female populations separately, I assumed a national adult sex distribution of roughly 48% men and 52% women, based on aggregate 1981 census figures. Using national literacy rates-97.5% for men and 91.6% for women in 1981 (Islami, 1982), and 98.8% and 95.2% in 1991 (Croatian Bureau of Statistics, 2005) - I computed the corresponding female-to-male illiteracy ratios. Each district's overall illiteracy was then decomposed into male and female components using these ratios and the assumed sex shares, from which literacy for each sex was derived as the complement. Finally, I applied proportional shrink-to-mean rescaling-adjusting district values by a constant factor so that their average exactly matches the national mean-before mapping the resulting male and female literacy rates.

Macedonia (1981 and 1991): I estimated sex-specific literacy rates from district-level data. Because district tables did not report male and female populations separately, I assumed a national adult sex distribution of roughly equal shares between women and men. For 1981, I used the female-to-male illiteracy ratio from the 1961 census as a proxy for the national gender gap, applying it to decompose each district's overall illiteracy into male and female components, which I then converted to literacy rates. For 1991, since an official national literacy mean was unavailable, I interpolated a national target between the 1981 and early-1990s benchmarks (Macrotrends, 2025) and used it as the reference level, repeating the same decomposition procedure. Finally, I applied proportional shrink-to-mean rescaling-adjusting district values by a constant factor so that their averages exactly matched the national means-before producing the male and female literacy and illiteracy estimates for mapping and analysis.

Montenegro (1981 and 1991): I estimated sex-specific literacy rates from district-level data. Because district tables did not include sex-specific counts, I used national female-to-male illiteracy ratios together with assumed adult sex shares to decompose each district's overall illiteracy into male and female components, which I then converted to literacy rates. For 1981, the adjustment was calibrated to match national literacy levels of 96.32% for men and 83.24% for women (CountryEcon-

omy.com, 1981), while for 1991 I applied the corresponding national ratio and targets. In both years, I used proportional shrink-to-mean rescaling-adjusting district values by a constant factor so that their averages matched the national means-before reporting sex-specific literacy and the complementary illiteracy rates.

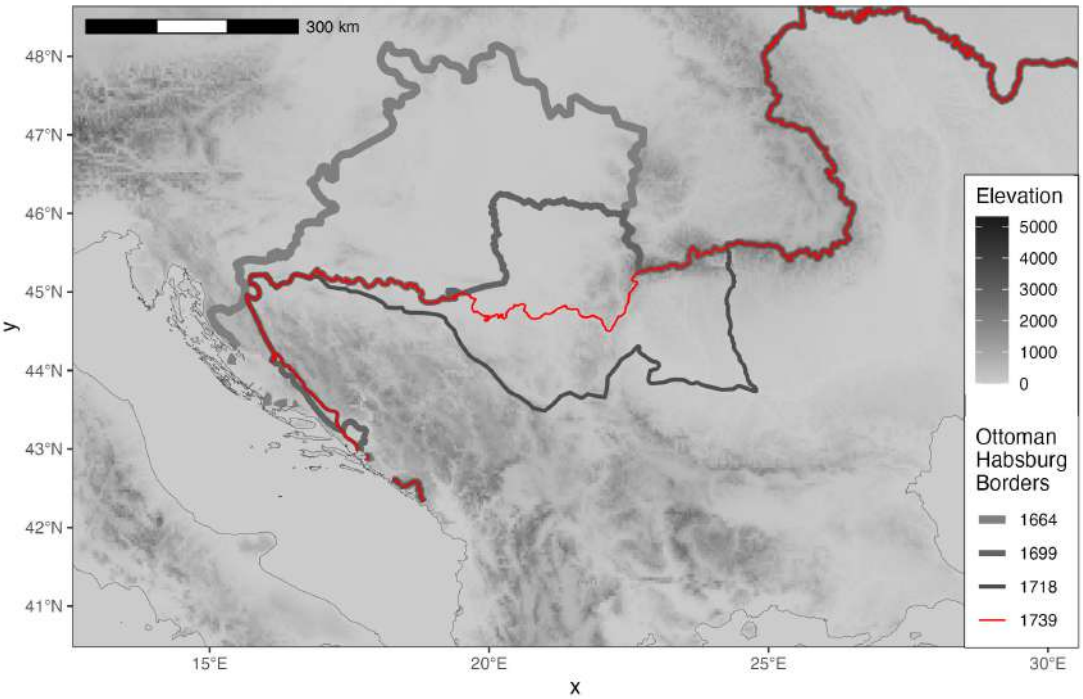
4 Drawing of the Border

4.1 The Ottoman-Habsburg Border(s)

One important question for the empirical strategy is how to deal with the shifting nature of the Ottoman-Habsburg border over time. Throughout the early modern period, various treaties (e.g., Karlowitz 1699, Passarowitz 1718, Belgrade 1739) redefined the frontier, with each demarcation persisting for different durations and carrying varying administrative consequences. To account for these temporal shifts, I focus on a subset of historically significant border years - 1664, 1699, 1718, and 1739 - that capture key turning points in Ottoman territorial control.

It is important to note that the 1648 Treaty of Westphalia is widely regarded as the origin of the modern state system based on fixed, sovereign borders (Abramson & Carter, 2016). The 1664 border marks the height of Ottoman territorial presence following the Treaty of Vasvár, whereas the 1699 Treaty of Karlowitz represents a decisive Ottoman retreat after the failed Siege of Vienna. The 1718 Treaty of Passarowitz reflects the peak of Habsburg gains into former Ottoman lands, and the 1739 Treaty of Belgrade marks a partial Ottoman recovery. All of these border configurations are displayed in figure A.14. This raises the important question of which border is most appropriate to use in a regression discontinuity design.

Figure A.14. *Borders*



4.2 Justifying the 1739 Border for Regression Discontinuity Design

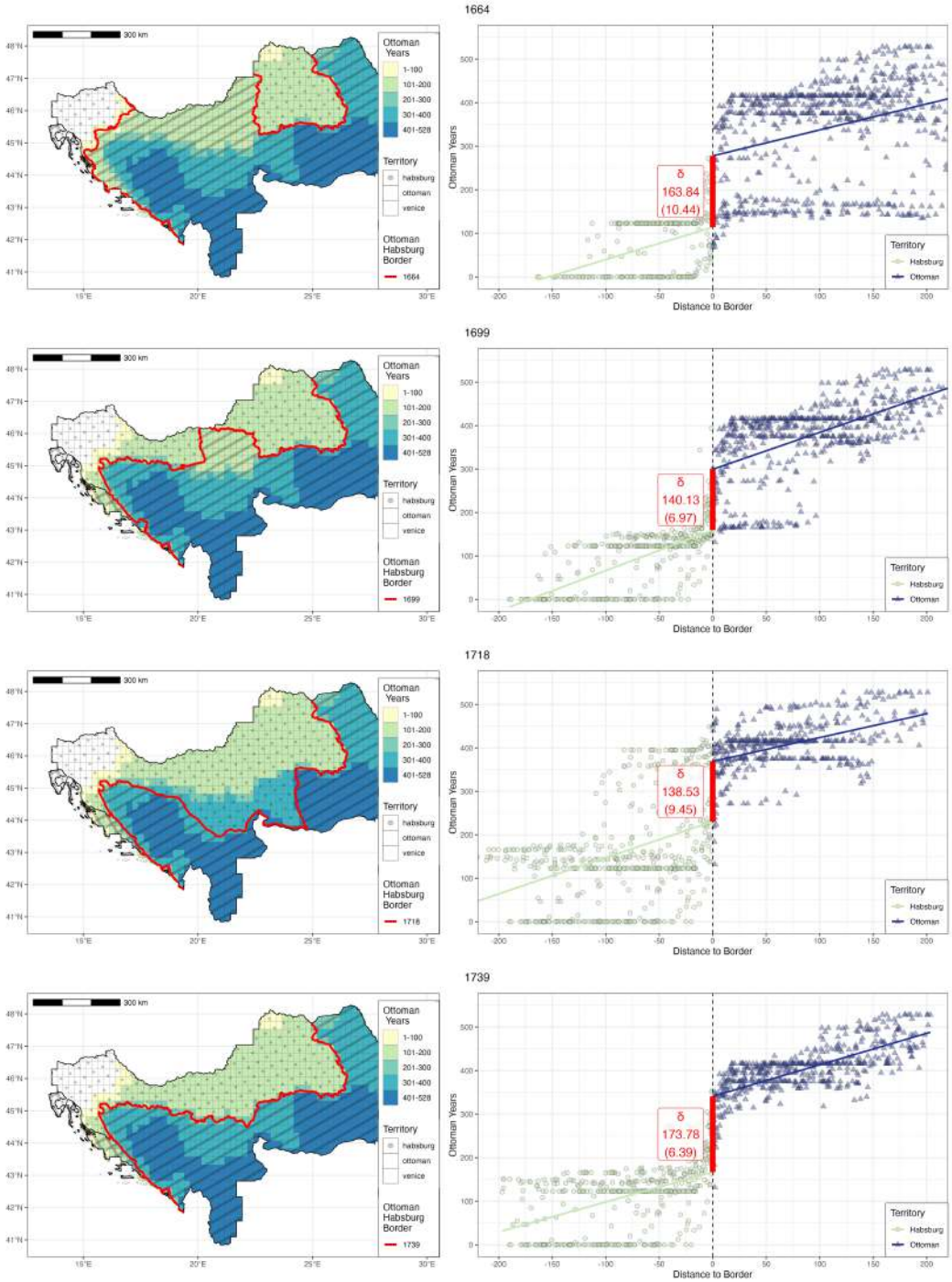
To justify the choice of the 1739 Ottoman-Habsburg border as the cutoff for the regression discontinuity design, I conduct a series of linear regressions for both the former Yugoslav and Romanian regions. As a first step, I dissolve the shapefiles for Romania, Serbia, Croatia, Bosnia and Herzegovina, Montenegro, Kosovo, and Slovenia into a single unified polygon. I then overlay this area with a regular grid of approximately 28 km (north–south) by 20 km (east–west) cells. For each grid cell, I calculate the duration of Ottoman rule using the historical shapefiles described in the previous section.

The border configurations examined correspond to the years 1664, 1699, 1718, and 1739. These dates are historically significant, each marking a major geopolitical reorganization between the two empires. The 1664 border followed the Austro-Turkish War and the Treaty of Vasvár, representing the height of Ottoman territorial reach in Central Europe. The 1699 Treaty of Karlowitz marked the first major territorial retreat of the Ottoman Empire following the failed Siege of Vienna. The 1718 Treaty of Passarowitz reflected the peak of Habsburg expansion into former Ottoman lands, incorporating northern Serbia and parts of Wallachia. The 1739 Treaty of Belgrade then re-established Ottoman control over much of Serbia, producing a relatively stable frontier that endured for several decades.

The results, presented in Figure A.15, indicate that the 1739 border yields the largest estimated discontinuity in Ottoman exposure (173.7 years). The reported jump, or δ, represents the predicted difference in years under Ottoman rule between locations immediately adjacent to either side of the border. Specifically, I estimate two separate linear regressions of Ottoman exposure on distance to the border - one for units on the Ottoman side and one for those on the Habsburg side - and compute the predicted values at the border point (distance = 0). The jump is calculated as the difference between these predicted values, and its standard error is derived from the squared sum of the standard errors of the respective predictions.

Earlier borders, such as those from 1664, 1699, and 1718, also produce substantial differences but explain less of the spatial variation in treatment intensity. These results support my claim that the 1739 border marks the most meaningful and enduring institutional divide and therefore provides the most appropriate setting for RD estimation. This conclusion is consistent with historical interpretations of the Treaty of Belgrade as establishing a long-standing and administratively significant frontier for much of the eighteenth century.

Figure A.15. *Borders*



4.3 The Making of the 1739 Border - The Treaty of Belgrade

Beyond identifying which border produces the largest discontinuities in exposure, it is equally important to understand how the line of demarcation itself was decided. The negotiations and boundary-making processes surrounding the Treaty of Belgrade (1739) shed light on whether the resulting frontier can plausibly be regarded as exogenous. This historical context is essential for evaluating one of the core assumptions of geographic regression discontinuity (GRD) designs-the as-if random assignment of treatment. Validating this assumption is crucial, as it determines whether GRDs can be interpreted as genuine natural experiments, in which "a natural or social process has assigned units to distinct conditions in a way that is as good as random" (Kocher & Monteiro, 2016, p. 952).

The credibility of GRD results hinges on the plausibility of the claim that the placement of the Ottoman-Habsburg border was effectively random, and that the outcomes we seek to explain are not driven by unobserved imbalances between the territories on either side of the border. If the as-if randomization assumption does not hold, it becomes impossible to "distinguish the effects of causes from the effects of the conditions under which they operate" (Kocher & Monteiro, 2016, p. 952).

For example, Kocher & Monteiro (2016) challenge the validity of the as-if random assumption in Ferwada & Miller (2014), who estimate the causal effect of delegating authority to the Vichy government on violent resistance during WWII. Kocher and Monteiro argue that the placement of the Line of Demarcation was not random but strategically chosen by Nazi Germany to maintain control over vital railway lines. As a result, the occupied zone contained a higher density of double-track railways, making it a more likely target for sabotage. This undermines the claim that differences in resistance levels near the line reflect the causal effect of indirect rule.

Similarly, Paine *et al.* (2024) question the as-if random assumption in the context of African borders. They argue that the placement of borders during the Scramble for Africa was shaped by precolonial political organization and geographic features such as rivers and lakes. In other words, the location of borders was heavily influenced by preexisting conditions on the ground, rather than being arbitrarily drawn.

Taken together, these critiques may appear to cast doubt on the continued viability of GRD designs. However, both Kocher & Monteiro (2016) and Paine *et al.* (2024) ultimately defend the use of GRDs, arguing that - when supported by detailed historical analysis and grounded in strong "qualitative historical knowledge" - GRDs can still yield meaningful causal insights. Moreover, Paine *et al.* (2024) make the distinct claim that, in cases where borders were drawn along geographic features such as rivers, "a claim of conditional randomness may be justified." However, they emphasize that "making such an assessment requires consulting the history and features of the border" (Paine *et al.*, 2024, p. 17).

When consulting the historical record, Kocher & Monteiro (2016, p. 953) recom-

mend conducting idiographic research,[16] which involves tracing the specific causal processes by which units were assigned to treatment and control groups.[17] Three key factors should guide this description of the assignment process: (1) the information available to decision-makers at the time, (2) the incentives they faced, and (3) their institutional or material capacity to influence assignment. Thus we follow these questions in explaining how the Ottoman-Habsburg border was drawn under the 1699 Karlowitz and 1739 Belgrade treaties.

(1) What information was available to the negotiators?

Before the late seventeenth century, European frontiers were typically porous zones rather than clear lines of demarcation. What matters for the GRD is that the Habsburg–Ottoman boundary became a clearly delineated frontier after the Treaty of Karlowitz (1699), and in durable form after the Treaty of Belgrade (1739). Roughly 70% of the 1739 line coincided with the 1699 demarcation.[18] Joint commissions of engineers and cartographers conducted on-the-ground surveys to acquire detailed information about the physical, demographic, and economic features of the frontier. They documented the course of rivers, marshes, and ridgelines to identify defensible and stable natural boundaries, while also recording the locations of forts, villages, and trade routes. Surveyors noted soil quality, road conditions, and access to water. Rivers or mountain ridges became the anchor for the demarcation line (Fodor, 2009; Kurtaran, 2018; Gümüşçü, 2010). As Stoye writes, "in war, or if war threatened, the river became a military frontier adjoining a communication system for the forces" (Stoye, 1994, p. 83). Since mass literacy emerged only later, there is little reason to think negotiators had information about education outcomes when fixing the boundary.

(2) What incentives-military, political, or economic-shaped their choices?

The commissions' information-gathering was guided by pragmatic military, political, and economic considerations, consistent with the principle of *uti possidetis*, whereby each side sought to retain control over territories it effectively occupied at the close of hostilities (Ingrao, 2000; Rothenberg, 1966; Shaw & Shaw, 1976; Stoye, 1994). Militarily, both empires prioritized defensible terrain-rivers, ridgelines, and marshes that could serve as natural fortifications and reduce the need for costly garrisons. Politically, they aimed to establish a clearly demarcated and mutually recognized border to minimize local skirmishes and stabilize diplomatic relations. Economically, surveyors paid particular attention to trade corridors along the Danube, Sava, and Drava rivers, ensuring that critical commercial routes and customs revenues remained under their respective control. The resulting boundary thus re-

[16] This is similar to what Dunning (2012, pp. 212–218) calls "treatment-assignment causal process observations", or what Collier (2011) refers to more broadly as a causal process observation.

[17] Both Kocher & Monteiro (2016, p. 953) and Dunning (2012, pp. 212–218) also emphasize the importance of explaining (ii) how the treatment was administered, and (iii) how the causes connect with the observed outcomes.

[18] On the shift from jurisdictional to territorial sovereignty, see Febvre (1962). On Karlowitz, Belgrade, and the survey commissions, see Fodor (2009); Ingrao (2000); Rothenberg (1966); Shaw & Shaw (1976).

flected not only the balance of power on the ground but also a shared interest in creating a durable framework for defense, governance, and commerce.

Yet, the emphasis on defensibility and economic value was not applied uniformly along the frontier. While many rivers were chosen as convenient lines of demarcation, not all held equal strategic or commercial importance. As Stoye observes, the Una was "too narrow a stream, with too many fords" (Stoye, 1994, p. 91), while the "boundary along the Sava and its Slavonian hinterland interested him less. [...] A world of flood, forest and marsh faced hillier but otherwise similar country across the river. In Slavonia there were no fortresses, but no need for them." (Stoye, 1994, p. 186). These marginal stretches-defined more by geographic convenience than by deliberate military or economic logic-constitute the most credible quasi-experimental settings for assessing imperial legacies. In these areas, geography rather than politics determined exposure to empire, generating precisely the kind of local variation that permits causal inference about the long-term developmental imprint of Ottoman versus Habsburg rule.

(3) Did negotiators have the capacity to shift the line toward or away from particular communities?

Negotiators possessed some room to maneuver, but this capacity was constrained by coalition politics, logistics, and the need for enforceable borders. Deliberations often balanced symbolic parity with technical bargaining, and commissions implemented agreements by reconciling maps with terrain (Fodor, 2009; Stoye, 1994; Kurtaran, 2018). Where fortresses or trade routes were at stake, the line could be adjusted; elsewhere it followed natural features to avoid future tensions.

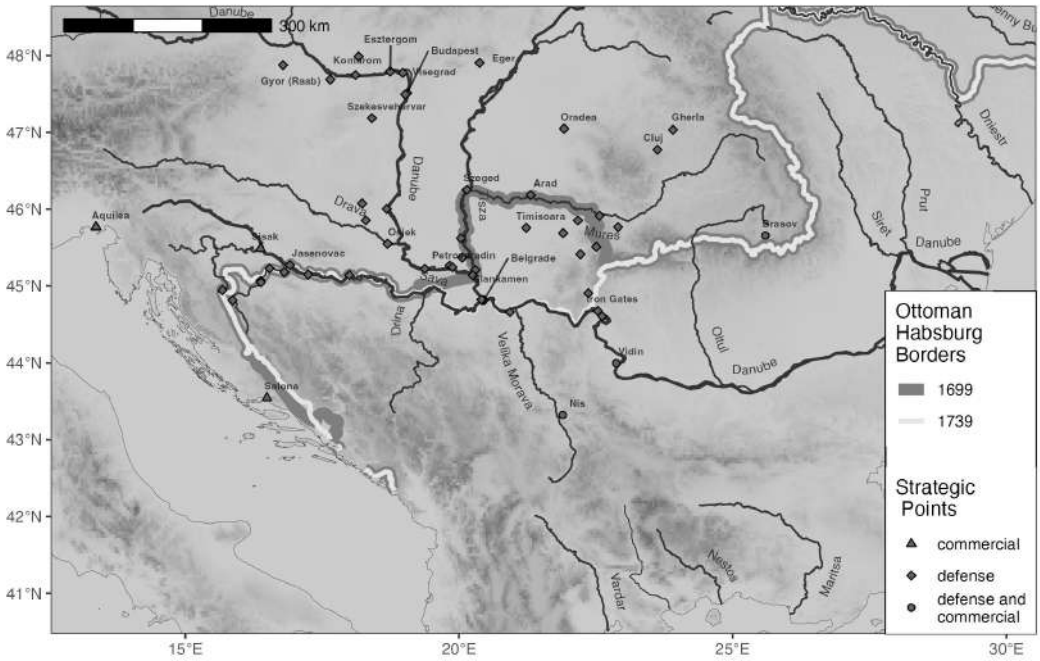
Taken together, the strategic priorities, economic interests, and diplomatic principles structured not just the logic of negotiation but also the concrete proposals for how the frontier should be drawn. The map in figure A.16 illustrates how these considerations translated into territorial arrangements, showing the key locations and rivers cited in historical sources alongside the final borders which overlap to a great extent decided at the Treaties of Karlowitz (1699) and Belgrade (1739).

Guided by these three criteria, I pre-specify primary GRD estimates on the Una and Sava segments-surveyed, militarily marginal, and exhibiting smooth geographic covariates across the cutoff-while reporting adjacent segments as robustness checks. This strategy privileges historical plausibility over blanket frontier coverage, thereby strengthening the credibility of the RD design.

5 RD2D Details Analysis

In the main text, I estimated heterogeneous treatment effects alongside the Ottoman-Habsburg border using the multi-score regression discontinuity (RD) design (Cattaneo *et al.*, 2025b,a). Specifically, the bivariate location-based estimation and inference procedure estimated treatment effects along the border using a location-score metric with data-driven bandwidth selection. The results distinguished between the

Figure A.16. *Strategic Points under Carlowitz Treaty*



portions that were drawn for geostrategic or economic reasons versus the ones that were drawn in an "as-if random" fashion.

Covariate Balance using Multi-Score RD design

In this section, I present an analysis of the many covariates along the border to justify the locations that are most credible for evaluating the impact of Ottoman rule on literacy. As indicated in figure A.18, elevation between the Ottoman and Habsburg areas varies considerably along the border. The average elevation for the whole Yugoslav and Romanian territory is 409 meters above sea level, with a standard deviation of 327 meters. Both the upper and lower panels for elevation indicate that elevation is sometimes higher and sometimes lower on the Ottoman side compared to the Habsburg side. In the west, near the coast in Croatia, elevation is higher on the Ottoman side. The difference between the two regions then shrinks along the rivers and becomes negligible in several places. In the Carpathians in Romania, the Ottoman side is generally at lower altitudes compared to the Habsburg side, although in the far north the Ottoman side again lies at substantially higher elevations.

Slope patterns are less pronounced. With a mean of 2.48 and a standard deviation of 1.97 across the territory, most of the estimated differences along the border are small and often not statistically distinguishable from zero. In Yugoslavia, slopes on the Ottoman side tend to be slightly lower or indistinguishable from those on the Habsburg side, while in Romania, some segments-particularly around indices 31 and 71-show the Ottoman side as noticeably steeper. Overall, slope differences

appear more localized and less systematic than elevation.

Precipitation differences are somewhat more systematic. The average annual precipitation across the territory is 62.71 cm with a standard deviation of 20.04. Along the Yugoslav border, the Ottoman side is consistently wetter in the western and central segments (7–16 cm higher), before converging to near parity further north. In Romania, the Ottoman side receives more precipitation in some segments (notably indices 31 and 71, with 6–7 cm more, in areas in the West), while in others the differences are negligible. Thus, while precipitation gaps emerge in several locations, particularly in western Yugoslavia, they are not uniform along the entire frontier.

Temperature shows variation but again without a consistent pattern. The average annual temperature across the territory is 9.57°C with a standard deviation of 1.81. Along the Yugoslav border, the Ottoman side is generally cooler in the southwest (differences around −2°C) but largely similar further north, with only minor positive or negative deviations. In Romania, the picture is mixed: the Ottoman side is warmer in several central segments (up to +2.6°C) but markedly cooler in the far north (−3.1°C). These fluctuations indicate that temperature differences, while sometimes sizable, do not follow a uniform east–west divide.

Finally, I examine wheat suitability, drawing on global FAO raster data on rainfed wheat cultivation potential. The mean suitability across the territory is 2.4 with a standard deviation of 1.01. Along the Yugoslav border, the Ottoman side appears more suitable for wheat cultivation in the western and central segments (differences between +0.4 and +1.4), though this advantage diminishes further east, with some negative or negligible values. In Romania, suitability is somewhat higher on the Ottoman side in several central locations (up to +0.9), but turns sharply negative in the far north (−2.4), where Habsburg territory had substantially greater agricultural potential. These patterns suggest that while parts of the Ottoman side were favorable to wheat cultivation, the advantage was uneven and even reversed in some regions.

Figure A.17. *Border Segment Covariate Balance*

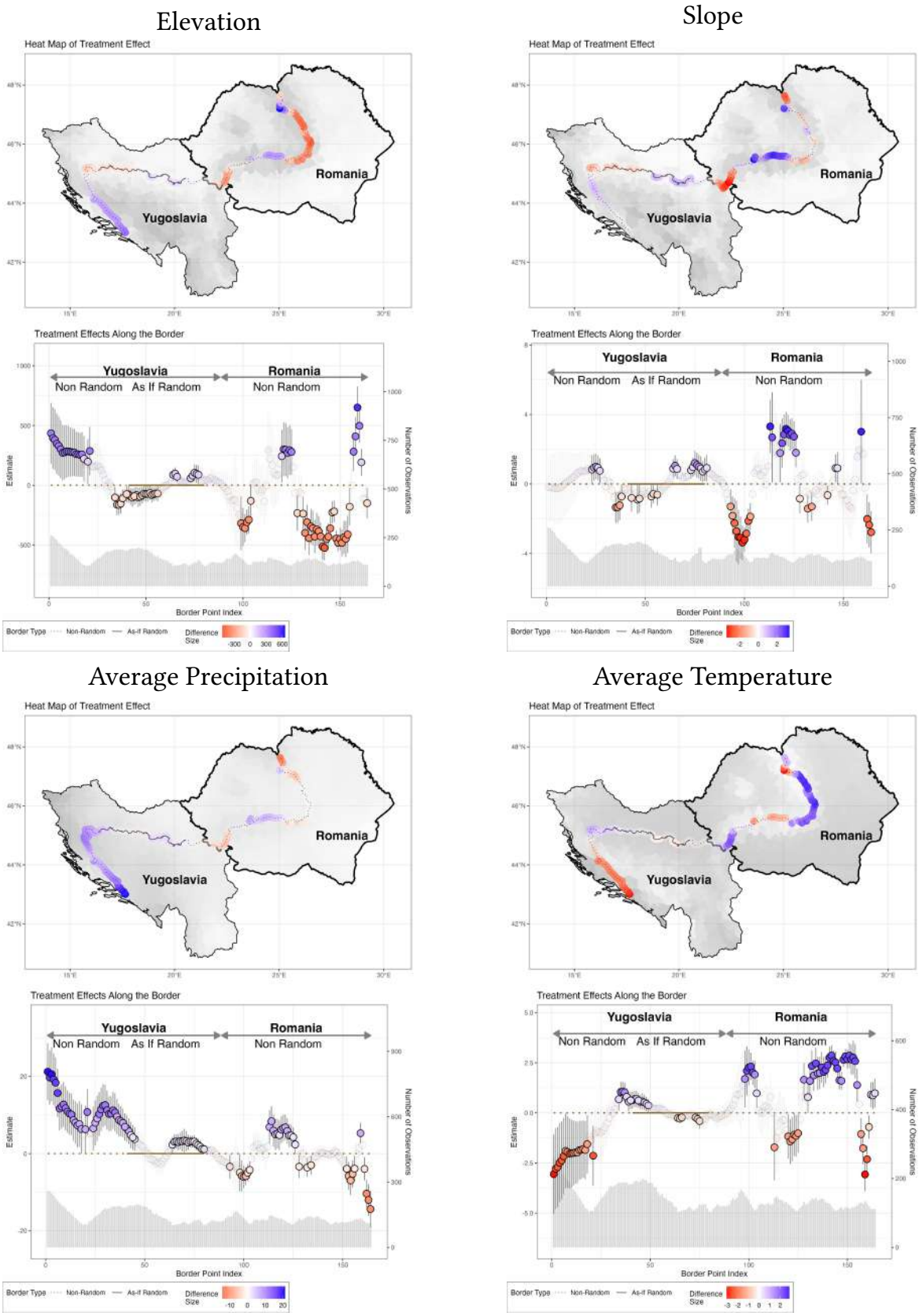
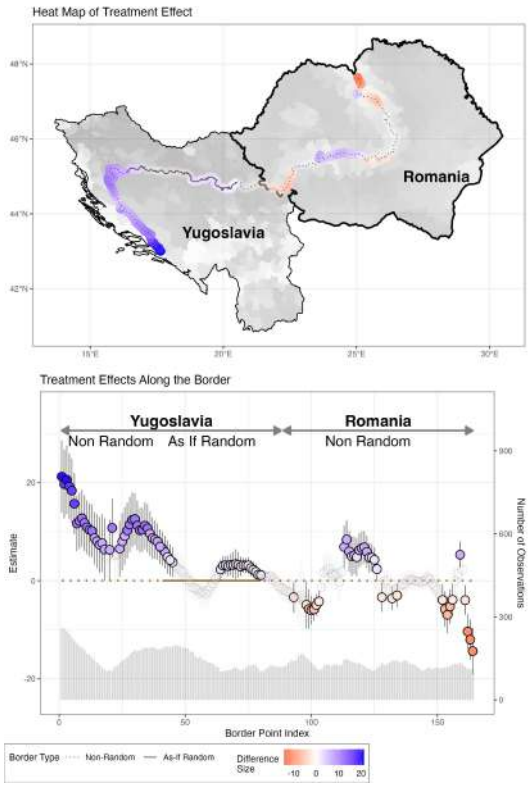


Figure A.18. *Border Segment Covariate Balance 22*

Wheat Suitability



6 Tests for Manipulation around the Threshold

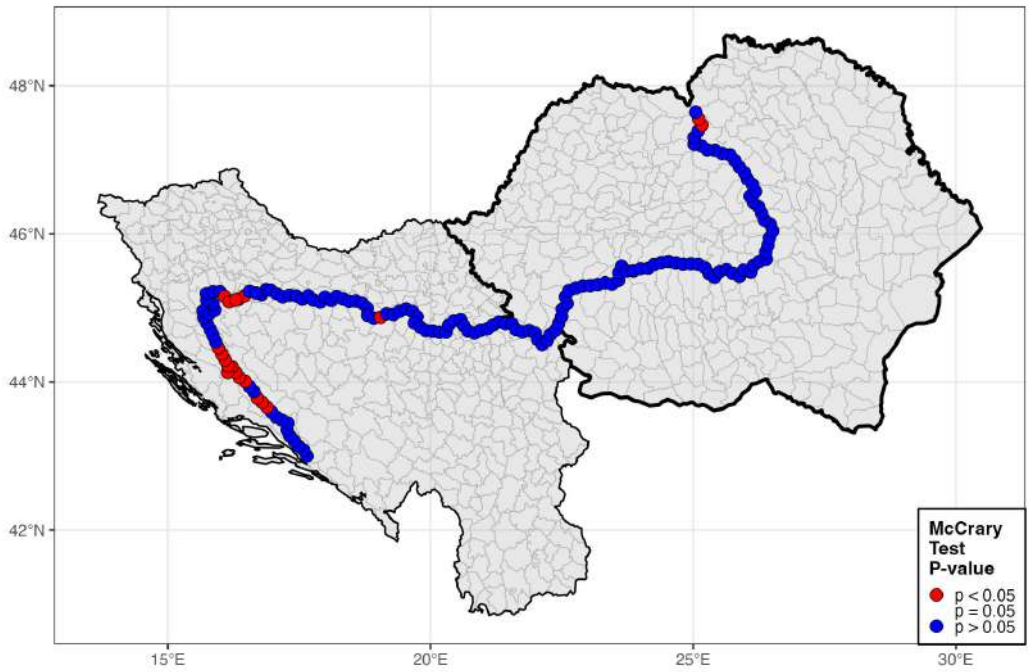
An additional identifying assumption is the absence of selective sorting across the treatment threshold. This assumption would be violated if people with different characteristics systematically moved from the Ottoman to the Habsburg side, thereby indirectly affecting literacy outcomes.

To test for such sorting, I employ the McCrary density test (McCrary, 2008), which detects discontinuities in the distribution of the running variable-in this case, distance to the border. A discontinuity would indicate potential manipulation, such as a higher population density on the Habsburg side due to migration.

While the traditional McCrary test is typically estimated once over the full sample, this approach may miss localized sorting along particular border segments. To account for this, I compute the test at 164 points along the border using the `rdplotdensity` function of Cattaneo *et al.* (2018).

Figure A.19 reports the resulting p-values, revealing significant discontinuities in the eastern portion of the border-particularly near the Dalmatian coast, the Una river, and the Croatia–Serbia boundary and two points in northern Romania. Consequently, discontinuity estimates at those border points should be interpreted with caution.

Figure A.19. *Density of Running Variable with Confidence Intervals*



7 Migration

7.1 Migration in Yugoslavia

An important question is the extent to which in- and out-migration might be impacting the results. The most important period that witnessed migration is 1930-1960. Political repression, economic conditions, and ethnic tensions drove outmigration among Yugoslavia's national minorities under communism. The most significant outflows occurred among Germans, Italians, Turks, and Albanians, while other groups, such as Hungarians and Romanians, also saw notable departures.

The German minority, once numbering around half a million, faced near-total expulsion after World War II. As early as 1943, the Communist-led Anti-Fascist Council of National Liberation "divested the Germans of all their rights and confiscated their land" (Shoup, 1963, p. 72). By 1963, fewer than 60,000 remained, with many forcibly relocated, deported to the Soviet Union, or fleeing to Germany and Austria. Estimates indicate that "183,000 remained unaccounted for" (*ibidem*).

The Italian minority, primarily in Istria and Rijeka, experienced mass migration after the redrawing of borders in 1945. Around 100,000 Italians emigrated between 1945 and 1948 (p. 80), driven by fears of persecution and economic instability. This exodus left "deserted and decaying fishing villages along Istria's western coast" (Shoup, 1963, p. 80).

The Turkish minority, mainly in Macedonia, also declined significantly due to voluntary migration facilitated by agreements between Yugoslavia and Turkey. Official records noted that "2,204 families had received permission to emigrate" by 1954 (Shoup, 1963, p. 80). Albanians (mostly Muslim) faced severe repression, particularly in Kosovo. Armed uprisings in 1944–45 led to prolonged conflict, and by 1945, "bitter fighting continued into spring, and martial law was not lifted until July" (Shoup, 1963, p. 71). Though not subject to forced expulsion, many left due to economic hardship and ongoing discrimination.

Unfortunately, there is no local level data to indicate these group changes between 1931 and 1961. To make up for that, I calculate migration by investigating population changes from 1930 to 1960 in absolute numbers.

7.2 Absolute Differences in Population

The fact that the units in 1931 and 1961 are not the same poses a challenge. For example, the unit in 1931 is *srez* or county. The unit in 1961, 1981, and 1991 is *opština* or municipality. The units for 1931 and 1961 are visible in the top left maps of figures A.20 and A.21.

As a first step, I calculated population density at a *srez* level in 1931 and at an *opština* level in 1961 by dividing the total population of an administrative unit by its area (see the top left maps of figures A.20 and A.21). I then created grids of approximately 24 by 24 km to make the censuses from 1931 more comparable to the one in 1961. Each administrative unit was then overlaid onto a regular grid, and the area

of intersection (A_{ij}) between an administrative unit i and a grid cell j was computed (see top right maps in figures A.20 and A.21). The proportion of the administrative unit's area intersecting each grid cell was used to distribute population accordingly (see bottom left maps of figures A.20 and A.21):

$$P_{ij} = D_i \times A_{ij}$$

where P_{ij} represents the estimated population of grid cell j based on the portion of administrative unit i that overlaps with it. D_i represents the population density for administrative unit i. This procedure was applied separately for 1931 and 1961, ensuring that each grid cell contained a population estimate for both years. This method ensures that population estimates are spatially consistent despite changes in administrative boundaries, enabling a robust comparison of demographic trends over time. As a final step, we also scaled the population estimate by computing an adjustment factor from total population sums in the three datasets (Yugoslavia 1931, Yugoslavia 1961, and grid data), then applied it to adjust the grid population values (see the results in the bottom rght maps of figures A.20 and A.21).

To summarize, figures A.20 and A.21 show the four steps performed to calculate population size at the grid level in 1931 and 1961. The top-left maps show the original units (srez for 1931 and opština for 1961), while the top right maps display the grid intersected with the polygons corresponding to the srez for 1931 and opština for 1961, depicting the area of the intersecting polygons. The bottom left maps show the population density at the intersecting polygons level while the bottom right maps indicate the final scaled calculations of population size at a grid level.

As a final step, we calculated the difference in population between 1961 and 1931 at a grid level. The left map in figure A.22 shows the area with positive changes in darker colors and the areas with no change or negative change in lighter colors. We also delineate the areas with negative change in blue contours. These are visible in the right map of figure A.22. The blue grids are regions where there were fewer people in 1961 compared to 1931, suggesting a process of outmigration.

Figure A.20. *Population Grid Level 1931*

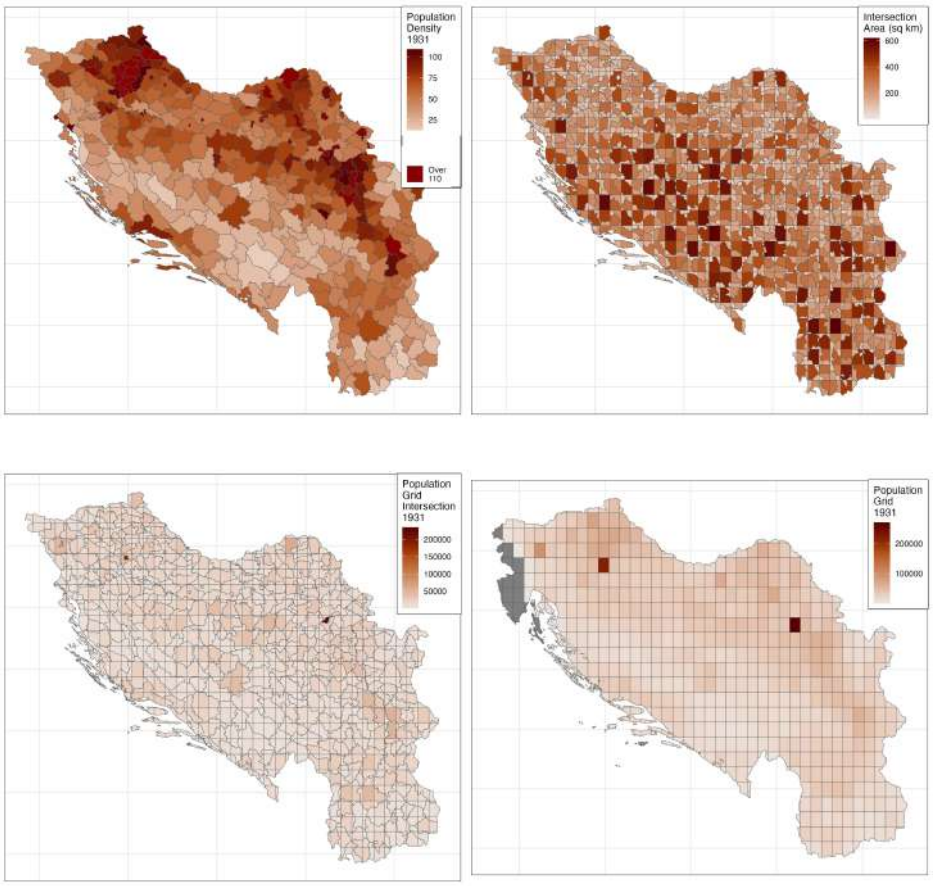


Figure A.21. *Population Grid Level 1961*

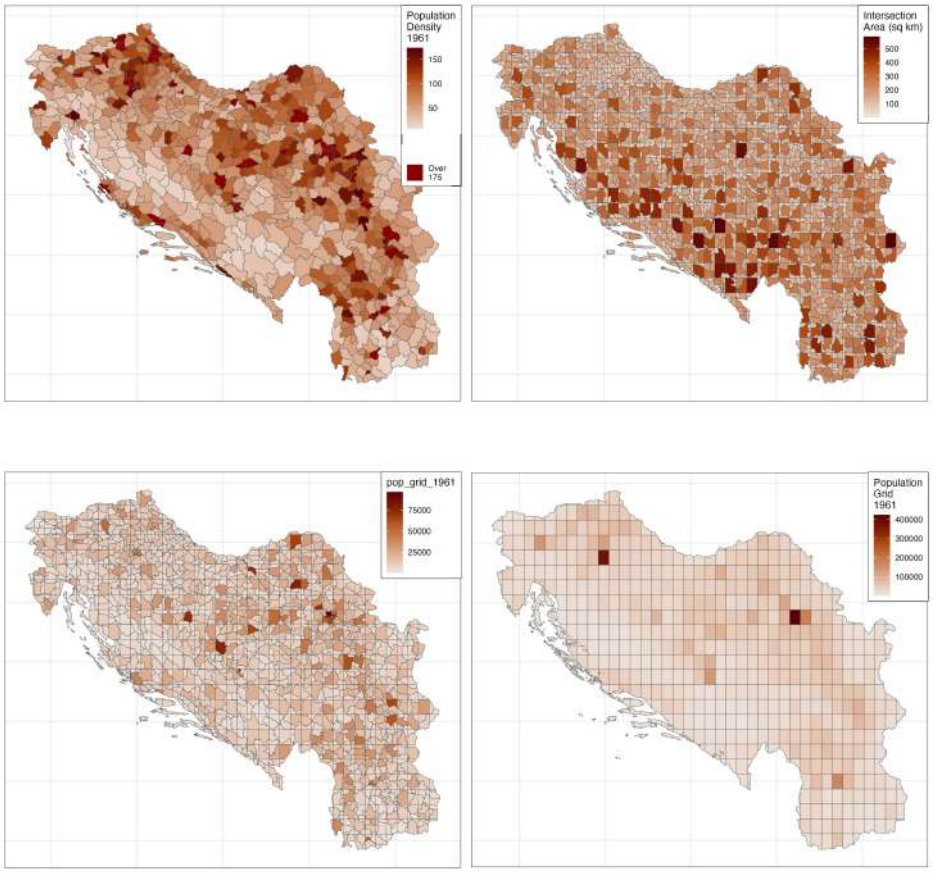
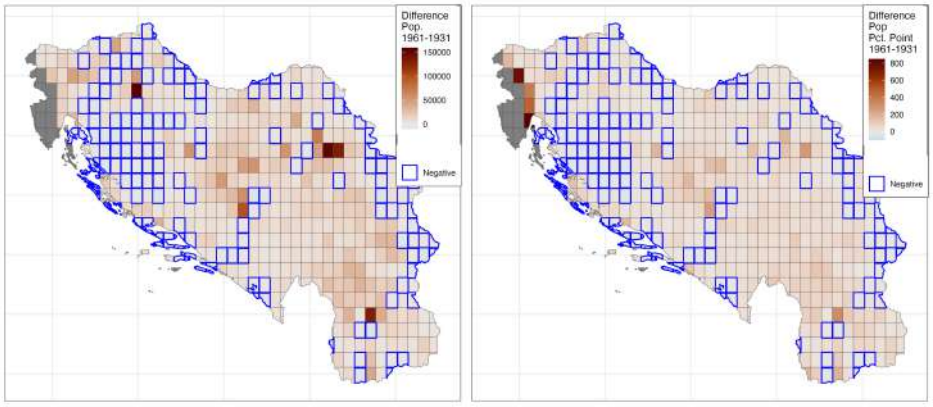


Figure A.22. *Population Changes 1961 - 1931*



7.3 *Migration by Religious Group*

I also calculated the percentage change in the three religions: Muslims, Catholics, and Orthodox. The percentage Muslims, Orthodox, and Catholics are clearly indicated in the 1931 census. For 1961, I added the number of Croatians, Slovenians, and Bulgarians, and then divided by the total population to calculate percent Catholics. To calculate percentage Orthodox in 1961, I added the number of Montenegrins, Macedonians, Serbians, and South-Slavs and divided by the total population. Finally, to calculate percentage Muslims in 1961, I added the number of Muslims and Albanians, and divided that number by the total population. The top maps in figures A.23, A.24, and A.25, show the rasterized percentage percentage of Muslims, Catholics, and Orthodox in 1931 and 1961. The bottom left map shows the rasterized difference in percentage points between 1961 and 1931, while the bottom right map depicts the zonal statistic at the 1961 *srez* level.

Figure A.23. *Difference Muslims in Percentage Points in Yugoslavia 1931-1961*

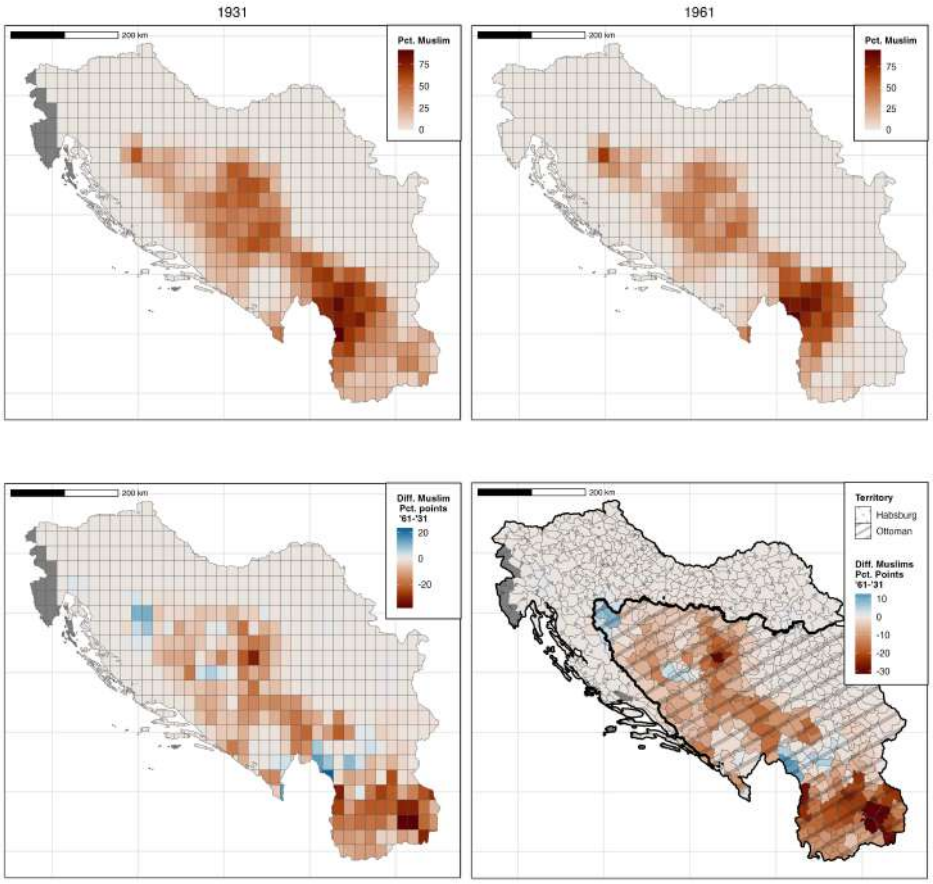


Figure A.24. *Difference Catholics in Percentage Points in Yugoslavia 1931-1961*

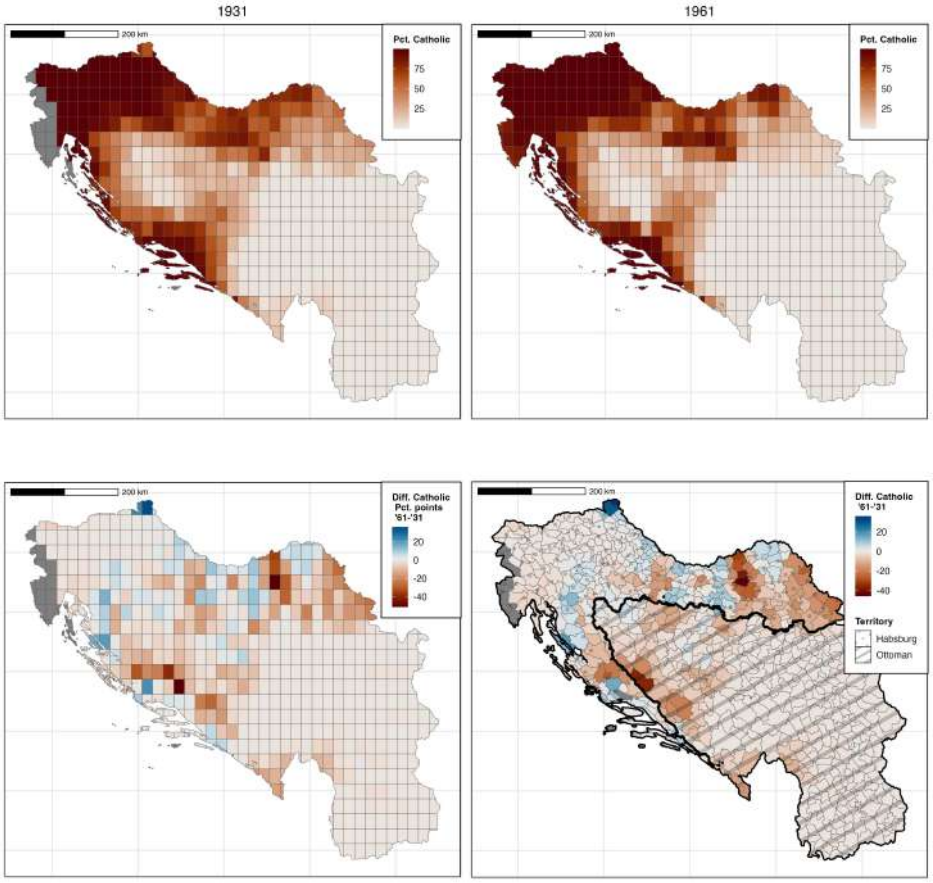
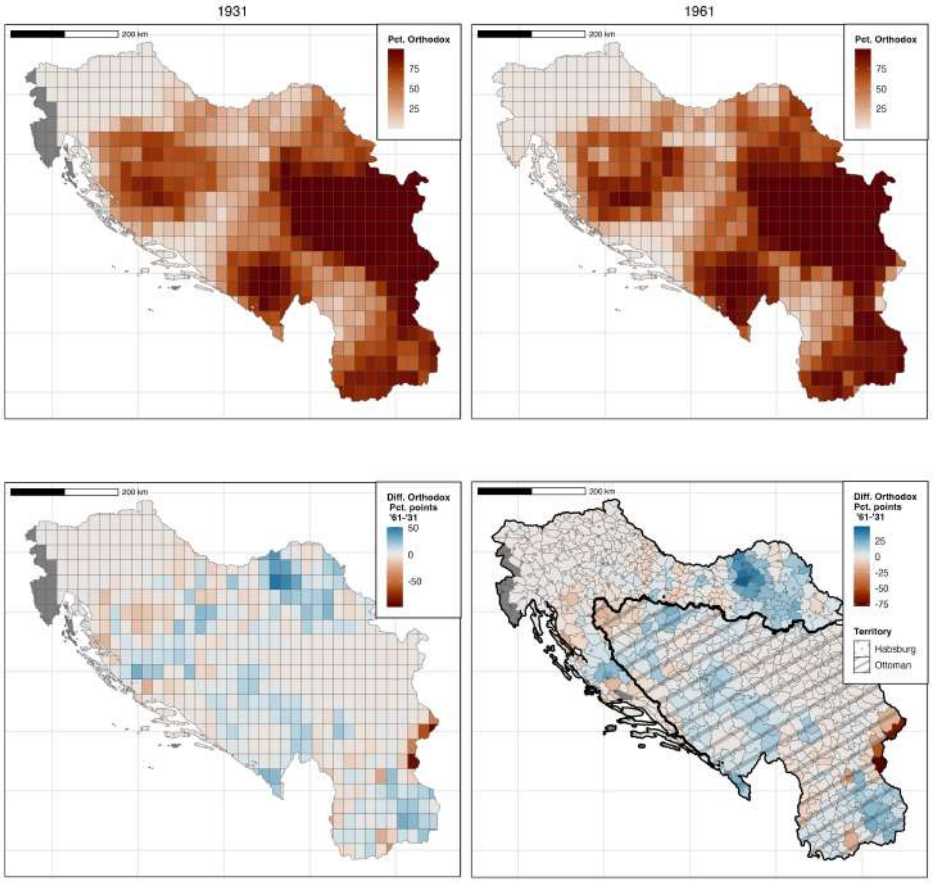


Figure A.25. *Difference Orthodox in Percentage Points in Yugoslavia 1931-1961*



7.4 Controlling for Migration

Given that some regions experienced migration, it is important to see how the evolution of literacy changes when controlling for migration. I thus compare the original model $Outcome_v = \alpha + \beta_{Ottoman}$ using the full sample and the non-parametric sample against four models: (1) the default model applied to a restricted sample, excluding units where the population in 1961 was smaller than in 1931; (2) the full sample with an additional control for absolute population change between 1931 and 1961; (3) the full sample with a control for percentage population change over the same period; and (4) the full sample with controls for percentage change in the Muslim, Orthodox, and Catholic populations between 1931 and 1961.

Figure A.27 shows all four of these models compared to the original specification depicted in gray. Across all specifications, the estimates remain highly consistent, suggesting that migration does not significantly alter our main findings.

7.5 Migration in Romania

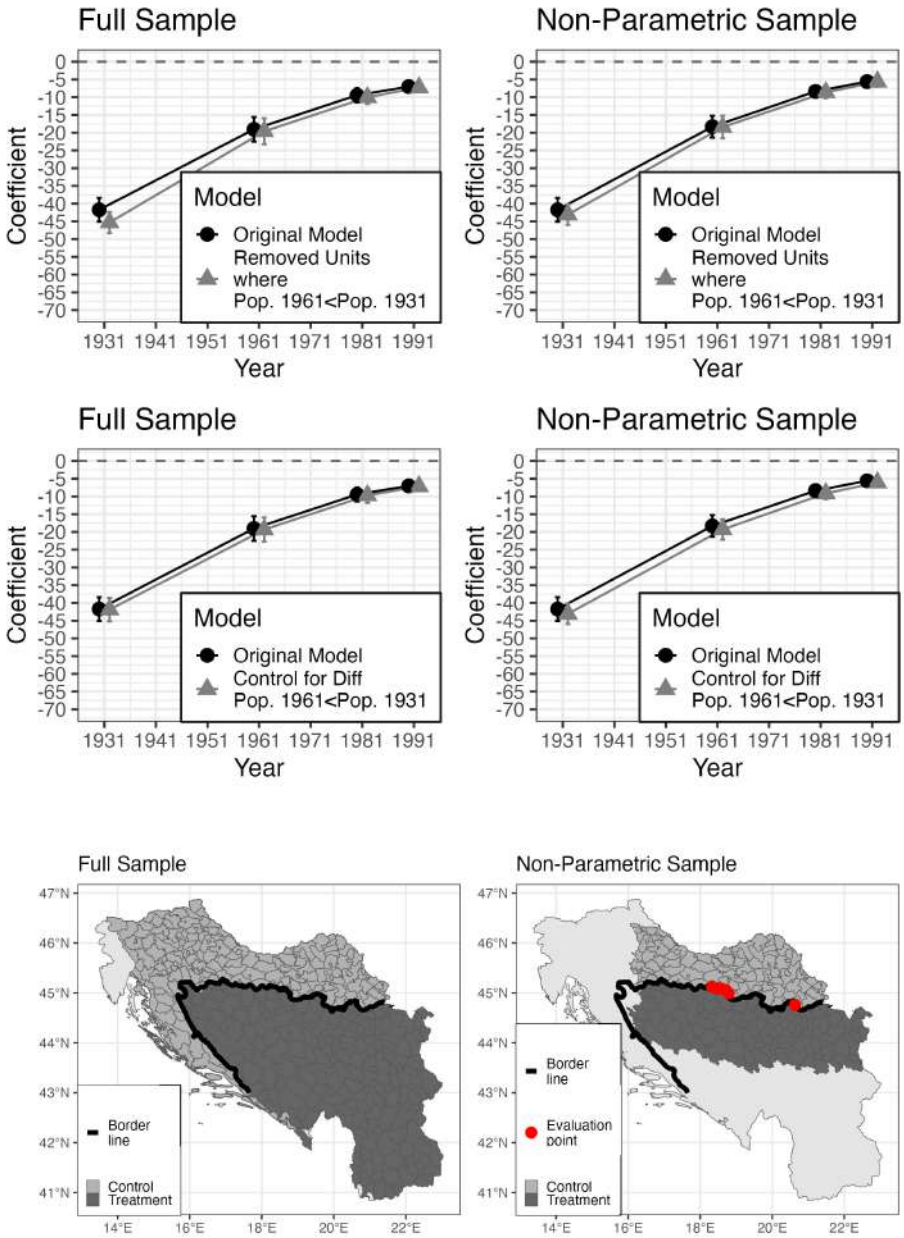
A related concern is whether migration might be driving the observed differences in literacy outcomes in Romania. Compared to Yugoslavia, however, large-scale population movements were considerably more limited. Romania did not experience mass expulsions or systematic ethnic cleansing in the interwar or immediate postwar period comparable to those affecting Germans, Italians, or Turks in Yugoslavia. While Romania did witness some emigration—primarily driven by economic factors and, to a lesser extent, political repression—these movements were smaller in scale and more spatially diffuse.

To assess the potential impact of migration, I rely on data on the so-called *populația flotantă* reported in the 1930 Romanian census (Institutul Central de Statistică, 1930). The floating population captures individuals who were temporarily absent from their locality of legal residence at the time of the census, typically due to seasonal labor migration, temporary urban employment, or short-term relocation. While this measure does not capture permanent out-migration, it provides the best available pre-treatment proxy for population mobility across regions in interwar Romania.

Importantly, although the census does not record later migration episodes, there are strong historical reasons to believe that large-scale population movements after 1930 were limited and unlikely to introduce systematic bias across regions. Unlike Yugoslavia, Romania did not experience major postwar border revisions, organized population transfers, or mass expulsions that would have disproportionately affected specific areas. As a result, differences in temporary mobility observed in 1930 are unlikely to mask substantial unobserved migration flows correlated with historical Ottoman rule.

In Romania, migration seems likely to have affected the results. I am using data on out-migration from Institutul Central de Statistică (1930). While the data only captures out-migration in 1931, it is unlikely that other migration episodes might

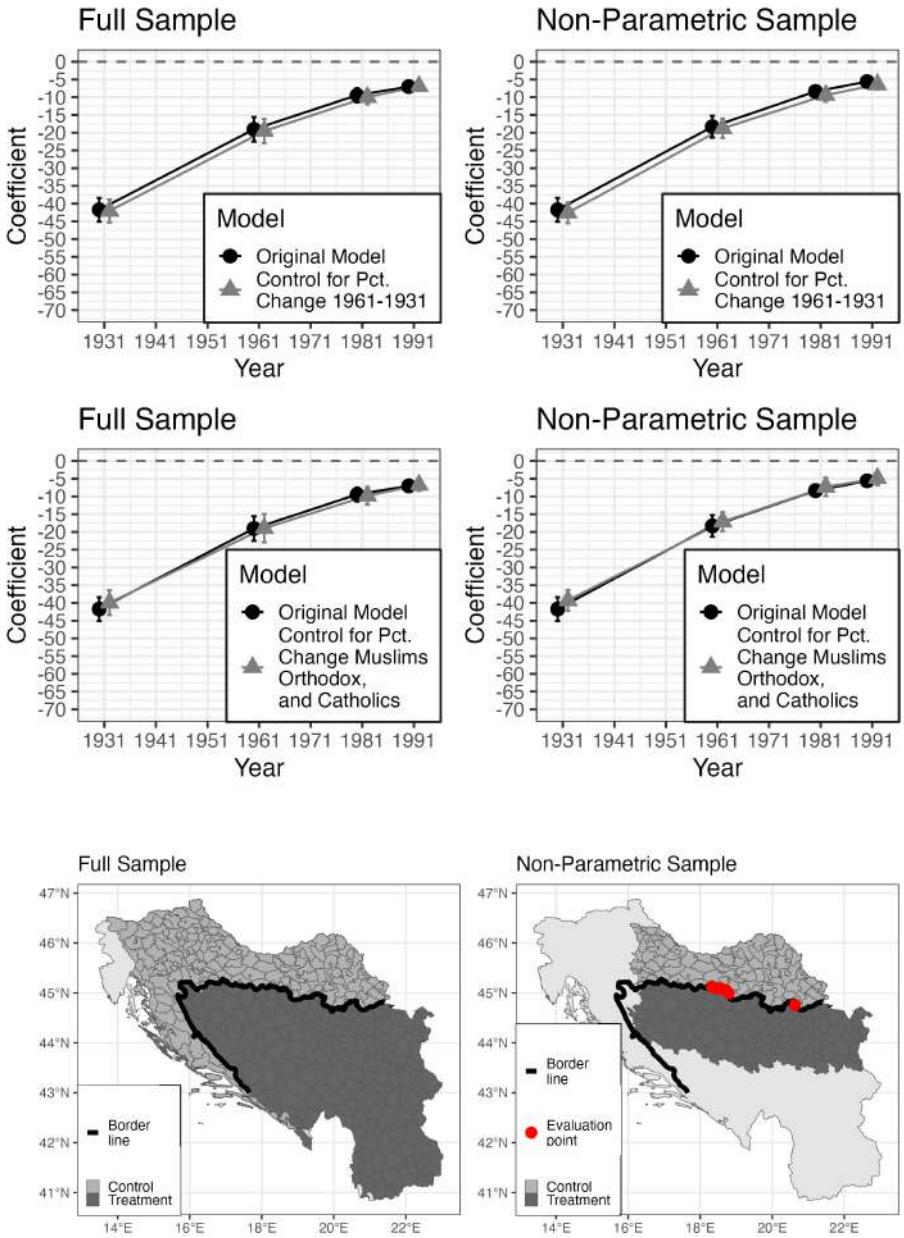
Figure A.26. Re-running the Analysis while Controlling for Migration



have affected the results, given Romania's history. To demonstrate that there are limited differences in out-migration between the two regions, I am regressing the treatment on out-migration based on data from 1930.

Rather than reconstructing population change over time, as was necessary in the Yugoslav case, I evaluate whether pre-treatment out-migration differed systematically across treated and untreated regions. Specifically, I regress the Ottoman treatment indicator on the level of out-migration measured at the local level in 1930. This approach directly tests whether areas historically exposed to Ottoman rule

Figure A.27. Re-running the Analysis while Controlling for Migration



were already experiencing differential population loss prior to the period of analysis.

The results are presented in Table ??. The estimated association between treatment status and out-migration is small in magnitude and statistically indistinguishable from zero. This suggests that out-migration patterns in Romania do not differ meaningfully between formerly Ottoman and non-Ottoman regions.

Table A.3

	Dependent variable:
	Pct. Outmigrants
Ottoman	0.027
	(0.048)
Observations	322
R^2	0.001
Adjusted R^2	−0.002
Note:	*p<0.1; **p<0.05; ***p<0.01

8 Federalism

A prominent alternative explanation is that Yugoslavia's federal structure weakened the center's capacity to equalize human capital across regions. On this view, decentralization constrained the implementation of uniform education policy, producing slower convergence in Yugoslavia than in unitary Romania. If federalism were the primary mechanism, convergence should be systematically slower across Yugoslav republics relative to Romania, holding constant exposure to the Ottoman legacy. This prediction is not borne out by the data.

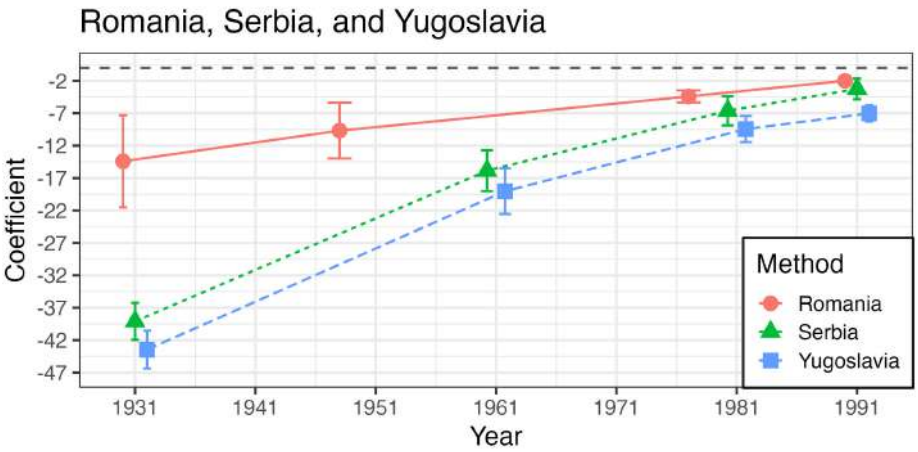
Two considerations motivate a focused test. First, Yugoslav federalism became more decentralized after the 1960s, particularly following the 1963 constitutional reforms and subsequent decentralizing changes that expanded republican autonomy. Yet the largest literacy gains and the steepest reductions in Ottoman–Habsburg differentials occur earlier, during the initial decades of socialist consolidation and mass schooling campaigns. The timing is therefore inconsistent with an account in which decentralization is the binding constraint on convergence: the period in which federal constraints should be weakest (early socialism) is precisely when convergence is most rapid.

Second, a within-Yugoslavia comparison provides a direct diagnostic for the federalism claim. I examine Serbia alone using the same non-parametric frontier-based sample employed in the main analysis. The logic is straightforward: Serbia was governed under the same federal constitution and thus faced the same decentralization constraints as Bosnia and Macedonia, and it operated under the same socialist education policies and national political economy as other Yugoslav republics. If federalism were the central impediment, Serbia should display convergence patterns similar to Yugoslavia as a whole. Instead, Serbia converges nearly as fully as unitary Romania. This test exploits the fact that federalism is constant across Serbian, Bosnian, and Macedonian districts, whereas ethnoreligious composition—and hence state–minority alignment—varies substantially. Also add Yugoslav federalism becomes more decentralized after the 1960s, yet most convergence happens before that,

I thus estimate the model $Outcome_v = \alpha + \beta_{Ottoman}$ using the non-parametric sample for Romania, Serbia, and Yugoslavia, with standard errors clustered at the county level. Figure A.28 plots the point estimates and 95% confidence intervals over time.

The results underscore the weakness of a federalism-based account. In Romania, the estimated Ottoman disadvantage declines from -14.4 percentage points in 1931 to - 2.0 points in 1991. In Serbia, despite operating within the Yugoslav federation, the Ottoman disadvantage falls from −39.1 −39.1 points in 1931 to −3.24 points in 1991. By contrast, for Yugoslavia as a whole the gap remains notably larger in 1991 (-7.01 points) and follows a slower convergence trajectory across the socialist period.

Figure A.28. *Federalism Test*



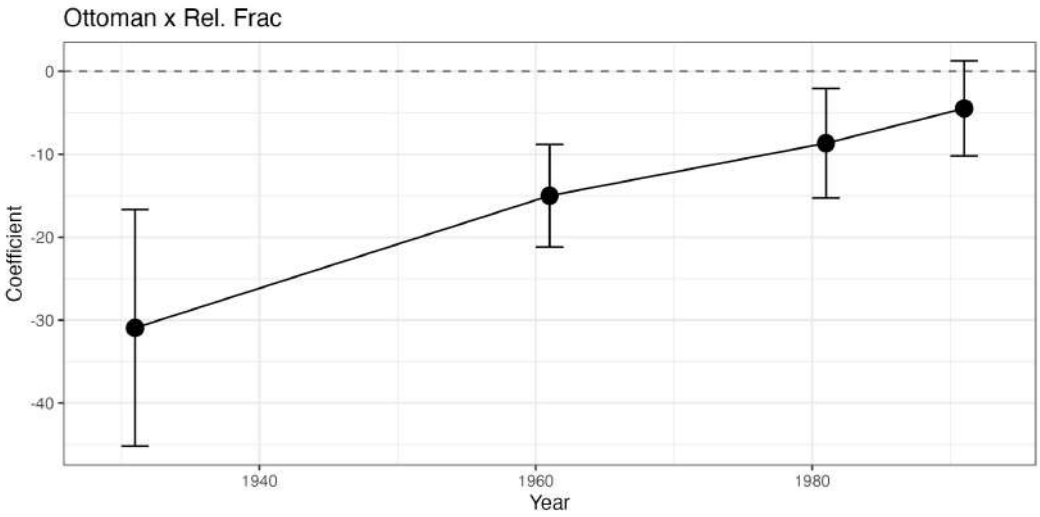
9 Differential Levels of Repression

Another prominent alternative explanation for why Romania experienced more complete convergence than Yugoslavia concerns differences in regime repression. One possibility is that Ceaușescu's Romania possessed a more coercive state apparatus than Tito's Yugoslavia, enabling more effective implementation of mass literacy campaigns through repression rather than consent.

I assess this explanation using two complementary tests that distinguish between repression as a general regime characteristic and repression as a mechanism that selectively enables convergence in specific social contexts.

First, I examine whether convergence occurred under Yugoslavia's own institutions in settings where state–society alignment was high, holding regime repression constant. I restrict the analysis to Serbian districts with less than 1% Muslims and more than 85% Orthodox Christians. These districts were governed under the same Yugoslav regime and exposed to the same education policies as Bosnia and Macedonia, but lacked the ethnoreligious misalignment emphasized in the theory. Figure A.29 shows that literacy converged rapidly in these high-alignment Serbian districts. This result demonstrates that Romania-level repression is not a necessary condition for convergence: even under Yugoslavia's comparatively less coercive regime, rapid convergence occurred where implementation frictions were low.

Figure A.29. *Literacy over time in Serbia in Homogeneous Districts*



Second, I conduct a placebo test using 2016 LITS data from Romania to examine whether individuals in the former Ottoman areas were more likely to report episodes of repression by the communist regime-either against themselves or their families. Specifically, I analyze responses to the following questions:

1) Did the government in Romania before 1989 engage in persecution, torture or any acts of violence against a) Yourself; b) Your immediate family (parents, siblings) and

2) While living under the pre-1989 government in Romania, were you or any members of your family sent to labor camp or prison for political reason: a) Yourself; b) Your immediate family; c) Your grandparents; d) Any other relatives.

The results, presented in Figure A.30, provide little support for the idea that individuals on the Ottoman side were more likely to have been repressed or exposed to coercive literacy campaigns than those on the Habsburg side. In other words, this placebo test rules out the possibility that targeted coercion toward Ottoman areas drove the within-Romania convergence.

Figure A.30. *Impact of Communist Repression on the Ottoman side in Romania*

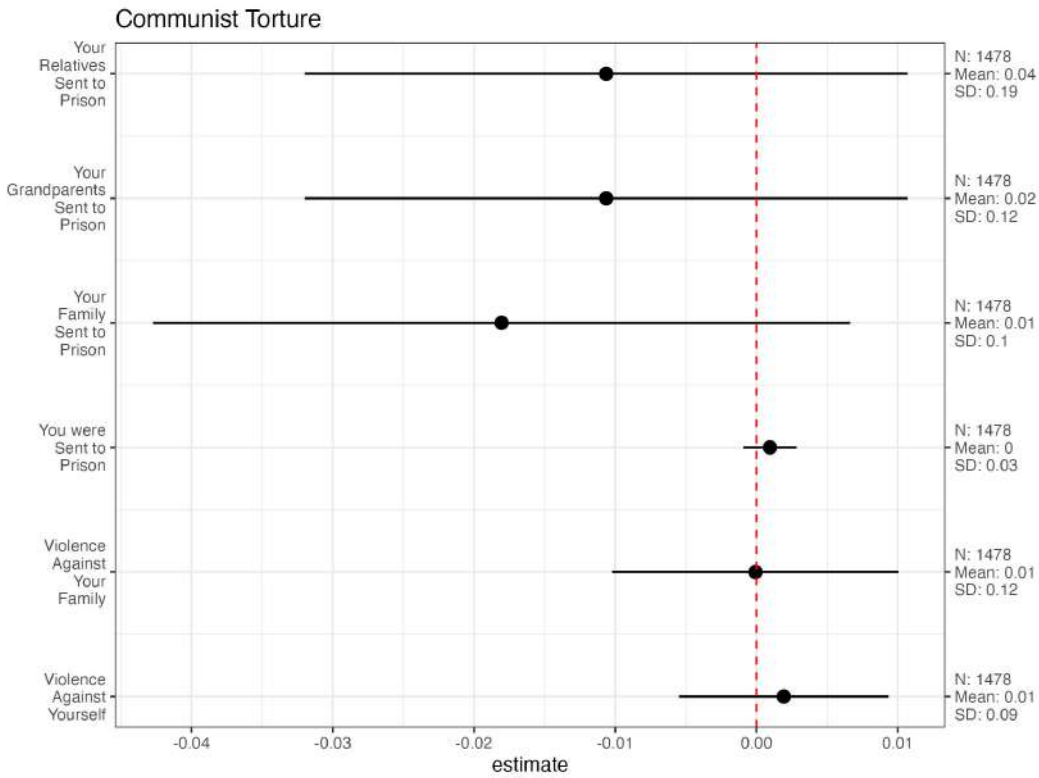
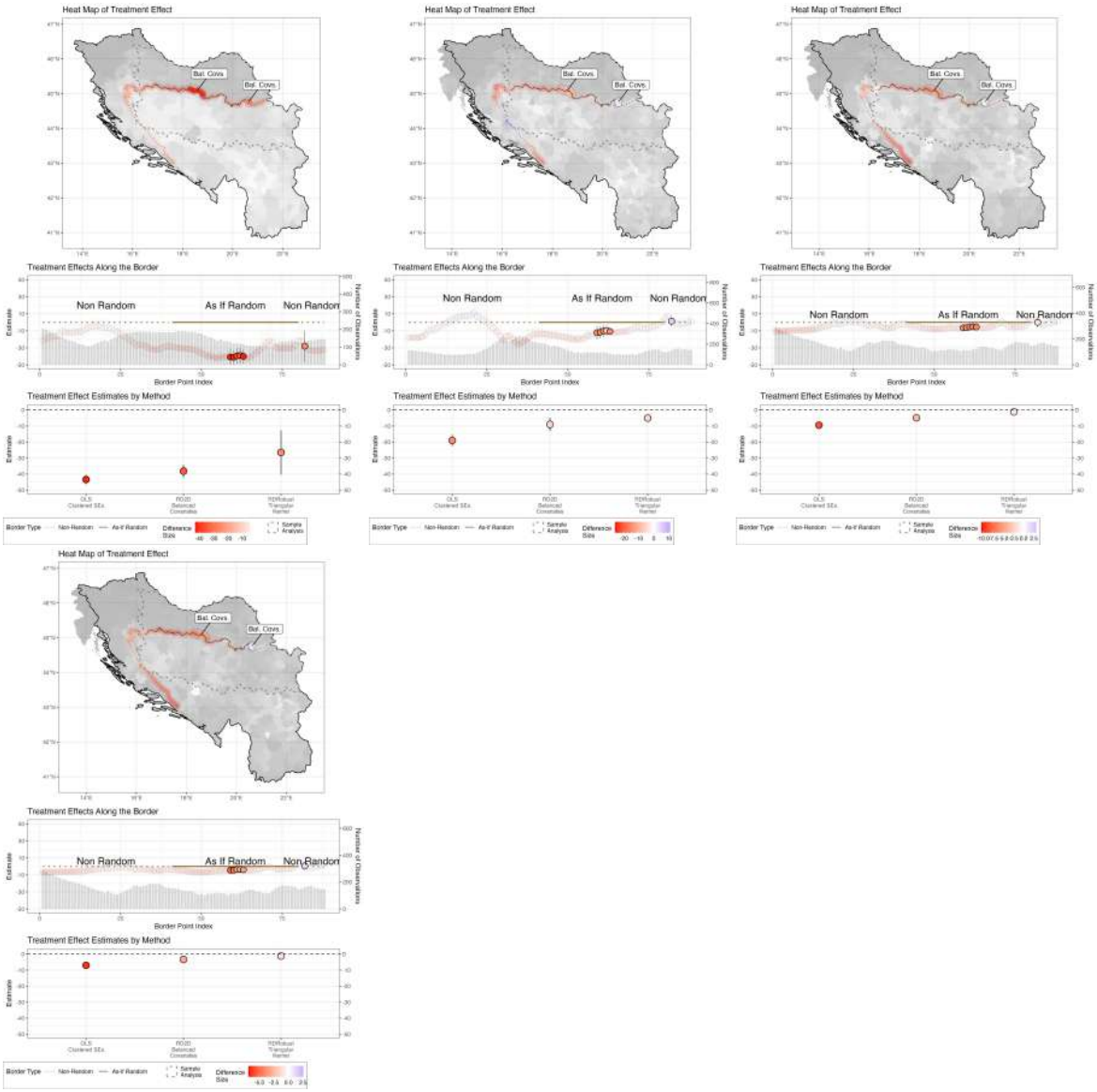


Figure A.31. *Evolution of Literacy over Time: 1931, 1961, 1981, and 1991 - Yugoslavia*



10 Multi-Score RD Design over time in Romania and Yugoslavia

Figure A.32. *Evolution of Literacy over Time: 1931, 1961, 1981, and 1991 - Romania*

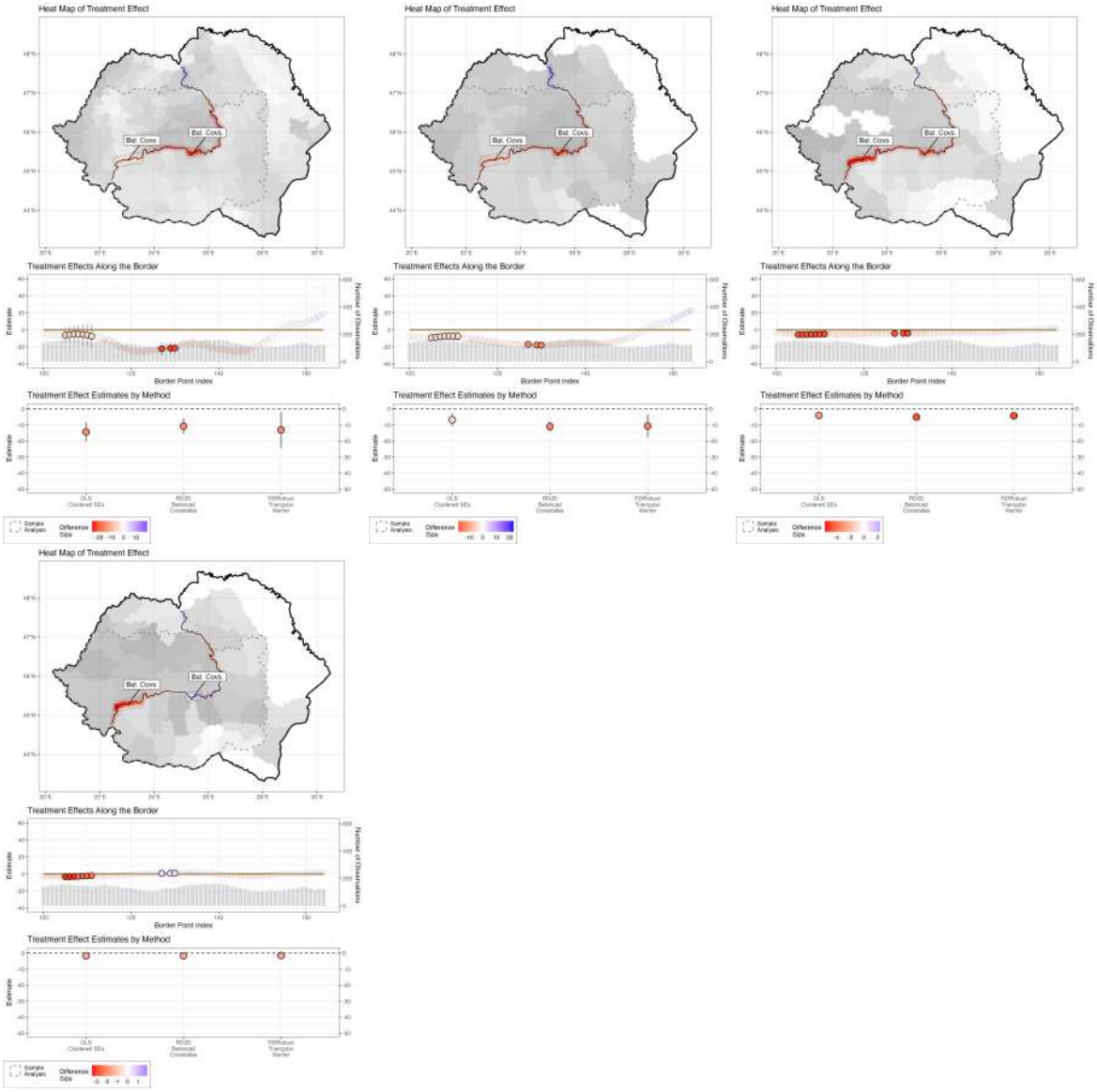
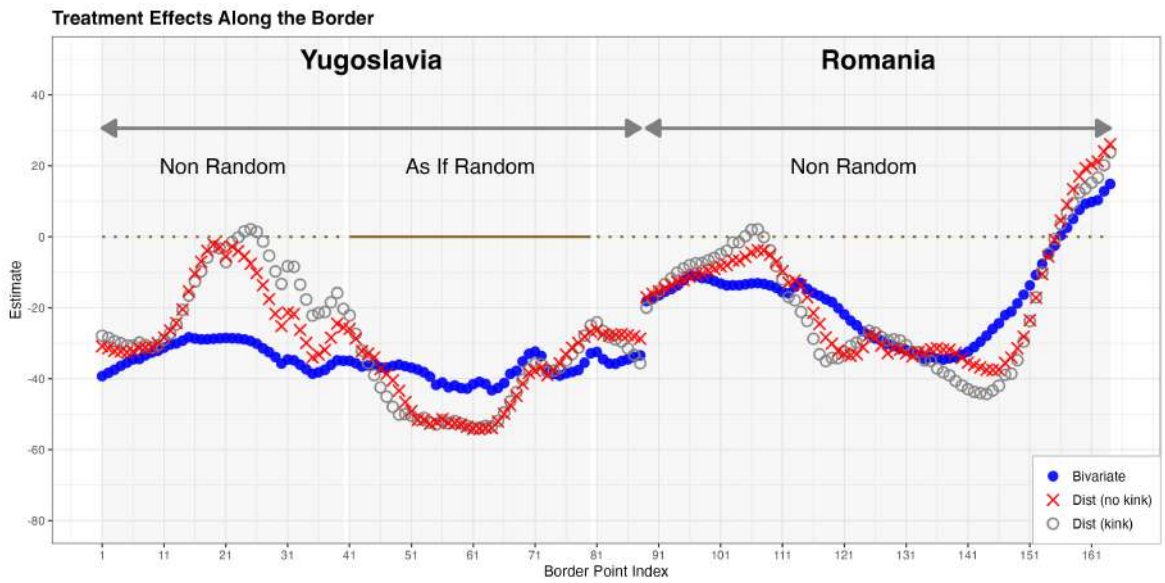


Figure A.33. *Border Segment Heteregenous Effects*



Multi-Score RD Design: Bivariate Location vs. Distance Approach

When it comes to the multi-score regression discontinuity (RD) design, Cattaneo *et al.* (2025a) explain that the approach based on the *bivariate location* relative to the boundary, presented in the main text, is an alternative to a *univariate distance* from the boundary. They show that the bivariate location method has both theoretical and practical advantages, which is why it is the main method presented in the main text. In particular, they explain that standard RD methods are not valid when distance is used, because features of the assignment boundary (e.g., smoothness, kinks, or irregularities) can generate substantial misspecification bias. This issue is especially problematic in geographic RD designs (Keele *et al.*, 2015). For this reason, Cattaneo *et al.* (2025b,a) recommend using the bivariate location score directly, together with guidance on bandwidth selection (Calonico & Farrell, 2020) and robust bias correction (Calonico *et al.*, 2014; Calonico, 2018).

Nevertheless, for completeness, I also report the results from the two distance-based estimations (with and without assuming the presence of border kinks), together with the bivariate regression discontinuity estimates. Figure A.33 therefore compares the two point-estimation approaches and demonstrates empirically the bias associated with the distance-based method. The black dots correspond to the treatment effect estimated using the data-driven MSE-optimal bandwidth. The crosses correspond to the distance-based treatment effect that ignores border kinks, while the empty circles represent the distance-based treatment effect that accounts for border kinks. As noted in the main text, 111 out of 164 points show statistically significant effects. Figure A.33 indicates substantial bias in the distance-based approaches (both with and without border kinks) along land sections of the border in Yugoslavia and Romania, and less bias along river sections of the border.

11 Traditional GRD

In this section I perform a standard geographic regression discontinuity analysis for completeness and comparability with prior work, since much of the existing literature has relied on distance-to-border designs. Including these results provides a benchmark against which the bivariate location-score approach in the main text can be evaluated. I present this analysis country by country: first Yugoslavia and then, Romania. Similar to the previous section, I examine the impact of Ottoman rule on 1930 literacy rates. The assumption that units close to the border are similar in observed and unobserved characteristics - aside from the imperial institution they inherited. The main model of interest can be written as:

$$\text{Literacy}_v = \beta_0 + \beta_1 \cdot \text{Ottoman}_v + f(\text{geographic location}_v) + \varepsilon_v \tag{3}$$

where Ottoman_v is an indicator variables equal to 1 if a settlement is inside the Ottoman area and 0 otherwise; f(geographic location is the RD polynomial, which controls for smooth functions of a district location in space. I run in the RD parametrically in a first instance.

I also estimate the RD model non-parametrically, following (Cattaneo *et al.*, 2019; Gelman & Imbens, 2017), where the function for geographic location is a local linear polynomial in the distance to the border. As Cattaneo *et al.* (2019) recommend, I use a triangular and uniform weighting kernel and calculate the optimal bandwidth bw using the MSE-minimizing procedure. I report them as a comparison here using the `rdrobust` estimation.

The primary running variable is the Euclidean distance to the Ottoman-Habsburg border. I assign treatment (treat = 1) to districts historically governed by the Ottoman Empire (using the territorial configuration established under the Treaty of Belgrade of 1739) and use the distance variable to define the running variable in both the non-parametric and parametric implementations.

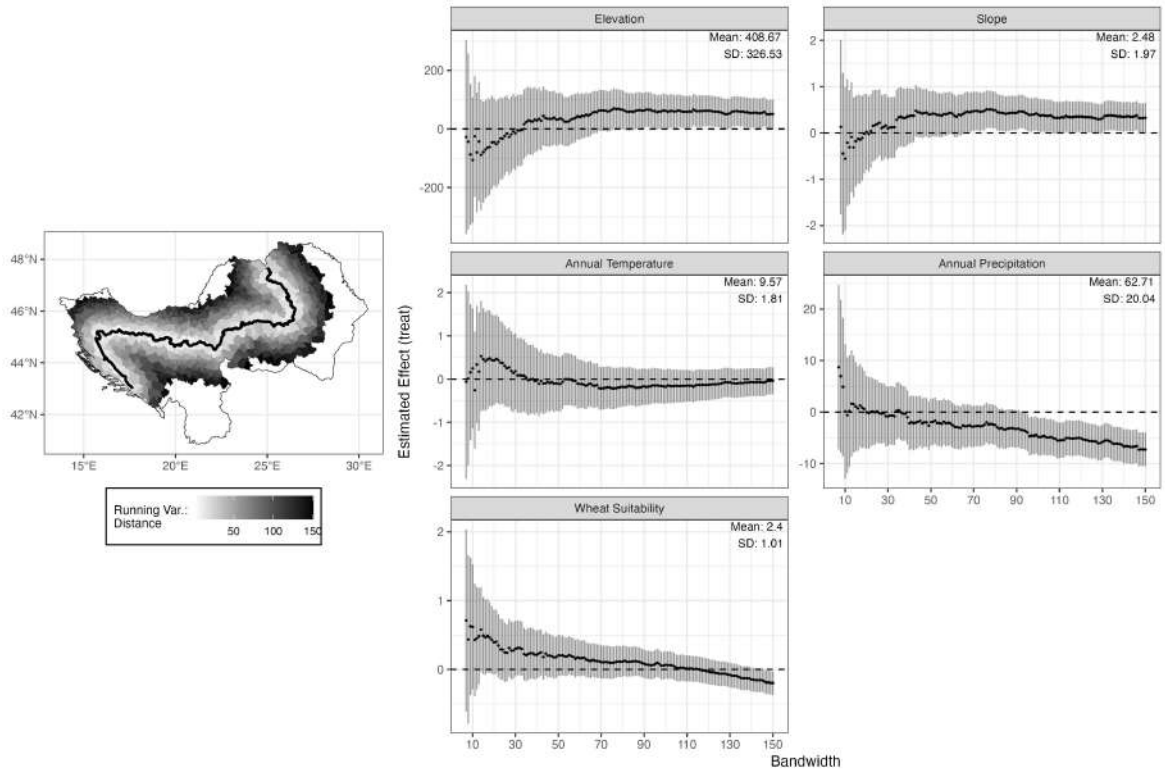
Covariate Balance in Traditional GRD

To assess covariate balance, I examine elevation, given the mountainous terrain in part of Yugoslavia and most parts in Romania (see the "Elevation" facet in Figure **??**). The average elevation across all districts within 150km is approximately 408 meters, with a standard deviation of 327 meters. Within narrow bandwidths around the Ottoman-Habsburg border, there is no statistically significant difference in elevation between treated and control districts: estimates are imprecise and close to zero. However, beginning around the 80 km bandwidth, treated districts appear to be consistently and significantly higher in elevation than control districts, with differences of roughly 50–65 meters ($p < 0.05$).

Mean slope-another key geographic feature-shows minimal imbalance in narrow bandwidths around the Ottoman-Habsburg border, with small and statistically insignificant differences in slope between treated and control districts up to 70 km.

However, beginning at around 70 km, treated districts consistently exhibit higher average slopes, with effect sizes increasing steadily and becoming statistically significant.

Climatic and agricultural covariates show mixed patterns of balance across the Ottoman-Habsburg border. Annual temperature exhibits no meaningful imbalance at any bandwidth: all coefficients remain small and statistically insignificant, suggesting that temperature is well balanced between treated and control areas. In contrast, annual precipitation begins to diverge significantly at broader bandwidths-beyond 90 km-where treated districts become consistently colder, with effect sizes growing to nearly 5°C lower by 140 km (note again that this results applies to a vast territory which includes both Yugoslavia and Romania). Wheat suitability is generally balanced both at narrower and wider bandwidths.



Traditional GRD Results

As a first step, I estimate a simple linear model using only the districts that are geographically adjacent to the Ottoman-Habsburg border, comparing literacy outcomes by treatment status. This specification offers the most direct comparison between treated and control units and minimizes differences in both observed and unobserved characteristics between the Ottoman and Habsburg areas. These effects are visible as diamond shapes in each facet of Figure **??**.

Beyond this localized comparison, I examine the robustness of treatment effects across functional forms and bandwidth choices. I estimate a series of parametric

linear models using bandwidths from 10 km to 150 km, in increments of 1 km. I begin with a simple specification that includes only the treatment indicator (see facet "Treat Only" of Figure **??**). Next, I incorporate region fixed effects[19] to account for broader regional variation (facet "Treat + Region FE"). I then add controls for geographic coordinates-latitude and longitude-to capture spatial trends (facet "Treat + Latitude + Longitude"). In subsequent specifications, I include the running variable itself (distance to the Ottoman border) - facet "Treat + Distance to Border," followed by a polynomial expansion of the geographic coordinates to allow for nonlinear spatial patterns (facet "Treat + Lat + Lon + Lat^2 + Lon^2 + Lat × Lon"). Additional models introduce a quadratic term for the distance to the border (facet "Treat + Distance + Distance^2"), and finally, I estimate a fully saturated model that includes a range of geographic and climatic covariates, such as elevation, slope, average temperature, precipitation, and wheat suitability (facet "Treat + Geography + Climate Covariates"). All models are estimated with standard errors clustered at the county level to account for spatial correlation within administrative units.

For each model and bandwidth, I extract treatment effect estimates, confidence intervals, p-values, and metadata (sample size, number of counties). I also record which covariates were dropped due to collinearity, to transparently document model behavior across varying sample windows. (some covariates are dropped in very close proximity to the border - around 10-15 km).

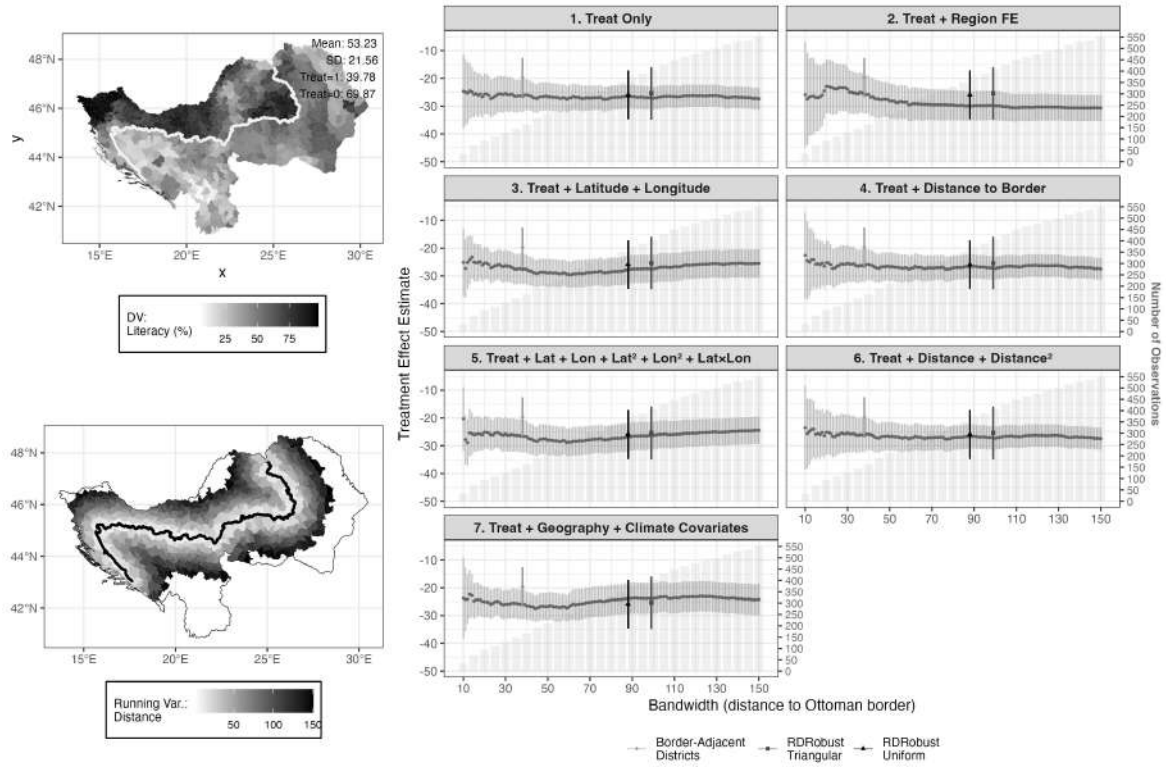
I also estimate two non-parametric models using the `rdrobust` estimator (Keele & Titiunik, 2015), which fits local polynomial regressions with MSE-optimal bandwidth selection. This approach allows for flexible estimation of treatment effects without imposing strong global parametric assumptions about the relationship between the running variable (distance to the border) and the outcome (literacy). Specifically, the estimator applies a simple polynomial functional form-but only locally around the cutoff-avoiding the risks of global model misspecification. For robustness, I use two kernel specifications: uniform and triangular (Calonico *et al.*, 2014). These non-parametric estimates are shown as squares and triangles overlaid on the parametric results in Figure **??**.

Results are visible alongside two maps that display the main dependent variable-percentage literacy in 1930 and the running variable - distance from the Ottoman-Habsburg border in increments of 1 km. Figure **??** also shows the results across models and bandwidths using: (1) point estimates and confidence intervals for treatment effects; (2) bars representing the number of observations at each bandwidth (3) overlays for the non-parametric RD estimates and adjacent-districts model. To ensure clarity, I differentiate models using both shades of grey and shape (triangle, square, diamond).

In terms of magnitude, the estimated effect of Ottoman rule on literacy in both Romania and Yugoslavia remains strongly negative and substantively large. In the

[19]The region fixed effects correspond to the banovinas in Yugoslavia in the early 1930s that take their name after the rivers present on their territory: Danube, Drava, Drina, Littoral, Morava, Sava, Vardar, Vrbas, and Zeta. In Romania, these regions are Transylvania, Wallachia, Moldavia, and Banat.

simplest comparison using only border-adjacent districts, treated areas exhibit an average literacy difference of approximately 25 percentage points. The two non-parametric `rdrobust` models yield similar estimates, ranging between −26.0 and −26.5 percentage points, reinforcing the robustness of the discontinuity at the imperial border. Parametric models stay relatively stable as one moves away from the border. This large gap remains stable across specifications that incorporate increasingly flexible spatial controls and covariates. Even in the most saturated model-which adjusts for regional fixed effects, geographic trends, and climate variables - the literacy difference remains substantial at −25 percentage points.



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